A STUDY ON THE PROBLEMS OF BUS TERMINALS IN DHAKA CITY: FUNCTIONAL ARRANGEMENT AND MANAGERIAL ISSUE

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A thesis submitted for the degree of

BACHELOR OF URBAN AND REGIONAL PLANNING

Date of submission: 17 April, 2019

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Abstract

Intercity bus terminals are critical nodes in national passenger transport, yet in many cities of the Global South they perform poorly despite repeated investment in their physical fabric. This study examines the three designated inter-district bus terminals of Dhaka—Sayedabad, Gabtoli and Mohakhali—through two linked lenses: their functional arrangement (the spatial organization of facilities and of bus and passenger flows) and their management (the actors, rules and power relations that govern their daily operation). Using a qualitative multiple-case design, the study combines field observation (November 2018–February 2019), ten key informant interviews with city corporation officials, bus owners' associations, labor unions and an academic expert, and three passenger focus group discussions. The functional arrangement of each terminal is benchmarked against the Bus Terminal Design Guidelines developed for India (SGArchitects, 2015), and stakeholder relations are analyzed using a power–influence matrix, an institutional Venn diagram, a historical timeline and cause–effect diagrams. The terminals share a consistent pattern of functional failure: passengers are loaded and unloaded on adjacent roads, ticket counters operate outside the terminals, the terminals double as depots and workshops, and passenger and crew amenities are inadequate or under-used. The stakeholder analysis shows that these outcomes are not explained by design deficiencies alone. Formal authority rests with the two city corporations, but effective control is exercised by politically connected bus owners' associations and labor unions, which govern access to routes, parking and revenue through informal arrangements. Chronic under-staffing, a stalled leasing contract and informal toll collection further weaken the public authority. The study therefore argues that the functional problems of Dhaka's bus terminals are, to a large extent, the spatial expression of an informal governance arrangement, and that physical upgrading or relocation is unlikely to succeed without institutional reform of terminal management, route permitting and revenue collection. The findings contribute to wider debates on urban informality and the governance of public transport infrastructure in rapidly growing cities.

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Chapter 1: Introduction

1.1 Background of the Study

Dhaka, the capital of Bangladesh, lies near the geographic center of the country and is the hub of its national road network, connected to six national highways (N1, N2, N3, N4, N5 and N8) (Google Map, 2019). Sustained rural–urban migration and the concentration of economic activity in the capital mean that a large and growing number of passengers enter and leave Dhaka by road every day, most of them by intercity bus.

Dhaka has three designated inter-district bus terminals—Sayedabad, Gabtoli and Mohakhali—which connect the city to the rest of the country via the national highways (DTCA, 2016). A bus terminal is the point at which a bus route starts or ends; buses use it to stop, turn or reverse and to wait before departing on their return journeys. A bus depot, by contrast, is the operating base of a transport system, providing parking, servicing and maintenance facilities, administrative functions and staff facilities; a fully enclosed depot is sometimes referred to as a garage (PPIAF, 2014).

Although terminals and depots serve distinct functions, all three terminals in Dhaka are used simultaneously as terminals and as depots. Demand for parking within them has grown steadily, while the space available has remained unchanged. Planning guidance assumes that an intercity terminal is located in or near the urban core and is directly served by local transit (Chiara & Crosbie, 1989); in Dhaka, terminals that were built at the edge of the city decades ago are now embedded in dense urban fabric and congested arterial roads.

Functional arrangement refers to the grouping and organization of resources according to similar activities or operations (Enna, 2015). In a bus terminal, it denotes the arrangement of facilities and elements according to the sequence of activities through which buses, passengers and staff move. The three terminals currently perform poorly, owing both to deficiencies in their functional arrangement and to weaknesses in their management. Yet few studies have examined how these terminals actually function and are managed, and none has asked how the two problems are related. This study addresses that gap by treating the terminals not only as physical infrastructure but also as sites of governance, where formal planning standards, statutory authorities and informal power relations meet.

1.2 Objectives and Research Questions

The study has two objectives:

- To examine the existing condition and functional arrangement of the inter-district bus terminals of Dhaka; and
- To identify the stakeholders involved in operating the terminals and analyze the managerial issues underlying the terminals' problems.

Together, these objectives address one overarching question: to what extent are the functional deficiencies of Dhaka's bus terminals a product of the way the terminals are governed? Three research questions guide the analysis:

RQ1. How do the existing facilities, and the flows of buses and passengers, in each terminal compare with a recognized standard of functional arrangement?

RQ2. Which formal and informal actors are involved in operating and managing the terminals, and how are formal power and actual influence distributed among them?

RQ3. Through which mechanisms do management and governance arrangements produce the observed functional deficiencies, and what reforms would address them?

1.3 Rationale of the Study

The inter-district terminals link Dhaka with the northern, southern, eastern and western regions of Bangladesh. Because the national highways converge on the capital, the performance of these terminals affects the national passenger transport system as a whole. Dhaka's concentration of employment and services draws large numbers of people from other cities and rural areas, and many of them enter the city through these terminals every day. The amenities provided are insufficient for this volume of passengers, several required facilities are absent, and the terminals do not function as intended despite the roles played by bus and terminal staff in their day-to-day operation.

Previous studies of Dhaka's terminals have documented their physical condition (e.g., Rahman et al., 2008), but none has examined their functional arrangement and management together. This study therefore focuses on the deficiencies in the terminals' functional arrangement, the roles of stakeholders in managing them, and the relationship between the two. Because Bangladesh has no national design standard for bus terminals, the functional arrangement of

each terminal is assessed against the Bus Terminal Design Guidelines developed for India (SGArchitects, 2015), a neighboring South Asian context with broadly similar bus operations. The question remains timely: in 2026 the Government announced plans to relocate Dhaka's inter-district terminals to the urban periphery, explicitly because they have become de facto depots and workshops (The Business Standard, 2026). The study can therefore inform both the design and, more importantly, the management of any future inter-district terminal.

1.4 Scope and Limitations of the Study

This study examines how the inter-district terminals of Dhaka function on a daily basis and who is responsible for operating and managing them. It documents the existing condition and functional arrangement of each terminal by tracing the flows of buses and passengers within and around it, and it examines the roles of stakeholders and the hierarchy of authority among them.

The study analyzes the actual, rather than the formally prescribed, process of terminal management and documents the issues faced by bus crews and terminal staff. It also assesses the availability and utilization of terminal facilities.

Finally, the study explores why the functional arrangement of the terminals is deficient and why the facilities provided are under-used, and it offers recommendations for improving both the functioning and the governance of the terminals.

The study faced several limitations:

- The master plans of the three terminals could not be obtained from the city corporations. Locating facilities precisely, and thus mapping the functional arrangement, therefore relied on field observation and the Detailed Area Plan (DAP).
- Because the terminals are largely managed through informal arrangements, many operational and managerial activities are undocumented and could only be reconstructed from interviews and observation.
- Passengers are often in a hurry when boarding or alighting, which made it difficult to recruit participants for focus group discussions.
- The time available for the study limited the depth with which some findings could be explored.
- Access to some stakeholders was difficult, and the time available for several key informant interviews was short.

- Observation could not be conducted during the Eid holidays, so conditions during peak festival demand are not captured.
- The study is qualitative and based on a small number of purposively selected informants. Its findings on revenue, staffing and informal payments rely largely on informants' accounts and could not be verified against official records.
- External government agencies (DMP, BRTA, RTC, DTCA and LGRD&C) were not interviewed; their roles are described from the accounts of internal stakeholders and from secondary sources.

1.5 Organization of the Study

Chapter 1 introduces the background of the research, its objectives and research questions, its rationale, and its scope and limitations.

Chapter 2 reviews the literature on the planning of bus terminals, stakeholder analysis, institutional frameworks for terminal management and urban informality, and sets out the analytical framework of the study.

Chapter 3 describes the research design, the selection of the study area, and the methods of data collection and analysis.

Chapter 4 profiles the study area and the three terminals.

Chapters 5 and 6 examine the existing condition and functional arrangement of the terminals.

Chapters 7 and 8 analyze the stakeholders and the managerial issues of the terminals.

Chapter 9 summarizes and discusses the major findings in relation to the literature and presents recommendations. Chapter 10 concludes the study and identifies directions for further research.

This chapter reviews the literature relevant to the planning, operation and governance of bus terminals. It first defines bus terminals and their types and sets out the facilities and functional relationships that planning guidance requires. It then reviews the concepts of stakeholder analysis, institutional frameworks for terminal management and urban informality, which together provide the lens through which the management of Dhaka's terminals is examined. Studies of bus terminals in Dhaka are few, and none has examined their management; the chapter therefore draws on international guidance and on the wider literature on transport governance in the Global South, and closes by setting out the analytical framework of the study.

2.1 Definition of bus terminal

A bus terminal is the point at which a bus route starts or ends. Buses stop, turn or reverse there and wait before departing on their return journeys, and passengers board and alight. A terminal also provides a convenient point from which bus services can be controlled. A depot, in contrast, is the operating base of a transport system, providing parking, servicing and maintenance facilities, administrative functions and staff facilities; a fully enclosed depot is sometimes called a garage (PPIAF, 2014).

A bus terminal is a designated area, separated from general traffic, in which buses and coaches can set down and pick up passengers safely and comfortably; ideally, it is located next to other transport terminals so as to provide the best interchange (Blow, 2005). Bus stops, terminals and depots are all critical infrastructure components of a bus-based transit system, but they differ in function. A bus stop is usually simply a place on the sidewalk where buses stop to pick up and set down passengers; a terminal occupies a larger area and handles the broader requirements of departing and arriving passengers; and a depot is where buses are parked and maintained.

According to SGArchitects (2015), bus terminals are nodal points at which journeys begin and end. They are important for their convergence function, serving as nodes of transfer between different modes.

2.2 Types of bus terminal

One of the primary determinants of a terminal's planning and design is the hierarchy of routes it serves (SGArchitects, 2015). Two broad types of terminal can be distinguished.

2.2.1 Intercity bus terminal

An intercity terminal is generally located in the downtown core and is directly accessible by local transit. It differs from other terminal types in that it serves long-haul services and handles a greater number of bus movements (Chiara & Crosbie, 1989; SGArchitects, 2015).

The routes served by such a terminal begin and end in different states, districts or cities, with long intervals between stops—usually more than 10 km (SGArchitects, 2015).

2.2.2 Local bus terminal – At a local bus terminal, routes begin and end at different places within the same city, and buses stop to pick up and set down passengers at short intervals, usually about 0.5 km (SGArchitects, 2015).

2.3 Essential requirements for planning a bus terminal

According to SGArchitects (2015), the essential requirements for planning and designing a bus terminal can be divided into primary infrastructure requirements and supporting infrastructure requirements.

2.3.1 Primary infrastructure requirements

Primary infrastructure, such as drop-off points, walkways and ticket counters, is provided in proportion to bus and passenger demand.

- **Facilities by user type**

Three user groups are considered in planning a terminal's primary infrastructure: passengers, terminal staff and bus staff. The facilities required for each are listed in Table 2.1 (Chiara & Crosbie, 1989; SGArchitects, 2015).

Table 2.1: List of Facilities in Bus Terminal

Passenger areas	**Areas for bus staff**	**Areas for terminal staff**
a. Ticketing and queuing b. Passenger waiting areas c. Passenger conveniences (drinking water facilities and toilets) d. Passenger circulation e. Boarding/Departing areas f. Facility entry g. Tourist information h. Security, including CCTV cameras i. Retail lease space j. Dormitories and lodging (if required) k. Cloak room	a. Canteen b. Resting areas c. Lodging areas (if required) d. Bus staff conveniences (drinking water facilities and toilets)	a. Revenue office b. Security and information c. Ticketing booth d. Resting room e. Staff conveniences (drinking water facilities and toilets) f. Canteen g. Maintenance staff h. Control room (CCTV surveillance)

- **Facilities for bus parking**

‘Layover’ refers to the area in which vehicles, having set down their passengers, are parked within the terminal until they are needed again. Its size should be based on parking requirements, and buses should be able to park without obstructing other bus movements. In some cases, depending on local timetables, spare bays can be used for layover (Hoque, 2011**;** Neufert et al., 2012).

According to Hoque (2011), where operators’ depots are some distance away, a canteen and toilets for bus crews should be provided at the terminal, so that crews do not need to return to the depot during breaks and between shifts. Where the depot is more remote still, the terminal should also provide locker rooms and ‘pay-in’ facilities. The ‘pay-in’ is an office where drivers and conductors hand over the fares they have collected, which are then checked and accounted for by clerical staff.

2.3.2 Supporting infrastructure

SGArchitects (2015) emphasize that providing the primary facilities is not sufficient for successful terminal planning: how these facilities serve users matters as much. Large and newly constructed terminals sometimes fail to meet commuters’ requirements and expectations because of the poor functionality of the facilities provided—dilapidated waiting areas, poor connectivity, shabby passenger sheds and unpleasant environments.

Supporting infrastructure—feeder infrastructure, seating, landscaping, lighting, signage and markings, public art and breakdown services—can enhance the user experience, efficiency and attractiveness of a terminal (Awino, 2014; SGArchitects, 2015). The supporting infrastructure that every terminal requires to operate effectively is described below.

- **Feeder infrastructure**

Feeder infrastructure enables other modes to access the terminal and act as feeders to it. It includes parking and drop-off and pick-up bays for private vehicles, taxis and rickshaws, and stops for local bus services (Chiara & Crosbie, 1989; Neufert et al., 2012). Such provision increases passenger convenience and intermodal accessibility.

- **Lighting**

Lighting should meet the required illumination levels and quality standards for both indoor and outdoor areas, and fixtures should be energy-efficient and require little maintenance (Awino, 2014; SGArchitects, 2015).

- **Signage**

Signage is an integral part of a terminal's wayfinding infrastructure and also helps to regulate vehicular and pedestrian movement. Signs provide the information and directions that ensure ease of access, convenience and passenger safety, and they should be easy to interpret (Awino, 2014; SGArchitects, 2015).

- **In-terminal breakdown services**

According to Hoque (2011), buses operating on long intercity routes often require minor repairs and maintenance to items such as fans, engine belts, tires and body panels. Terminals are usually not equipped for such work, because inspection, repair and servicing are the operator's responsibility and are carried out at a depot with a repair workshop and fueling, washing and garaging facilities (Awino, 2014; SGArchitects, 2015). Breakdowns nonetheless disrupt service schedules and inconvenience passengers, so provision for minor in-terminal breakdown services is important for efficient service planning.

Rahman et al. (2008) found that facilities at the Sayedabad, Mohakhali and Gabtoli terminals were insufficient relative to growing passenger demand, which they attributed to poor enforcement of scheduling regulations and poor maintenance. They reported an uneven distribution of facilities, a lack of information on arrivals and departures, illegal ticket counters, harassment of passengers by bus crews, and the poor condition of waiting rooms, toilets and other passenger facilities.

2.4 Functional arrangement of primary and supporting infrastructure

Functional arrangement is the grouping and organization of resources according to similar activities or operations, following their sequence (Enna, 2015). In a bus terminal, it refers to the arrangement of primary and supporting infrastructure according to the sequence of activities through which buses, passengers and staff move. A sound functional arrangement minimizes conflicts between vehicular and pedestrian flows and places each facility where its users need it (SGArchitects, 2015). The standard arrangement used as the benchmark in this study is presented in Section 6.1.

2.5 Stakeholders and stakeholder analysis

MacArthur (1997) defines stakeholders as individuals, groups, communities or institutions of any form, size and capacity that are involved in an organization, either internally or externally.

In the broader management literature, a stakeholder is any group or individual who can affect, or is affected by, the achievement of an organization's objectives (Freeman, 1984).

Internal, or primary, stakeholders are individuals or groups who participate in the management of the organization (AENEAS, 2009).

External, or secondary, stakeholders do not participate in the organization's day-to-day activities but are affected by its actions; they deal with the organization from outside and have little knowledge of its internal affairs (AENEAS, 2009).

Stakeholder analysis is an approach for generating knowledge about individuals and organizations in order to understand their behavior, intentions, interrelations and interests, and to assess their influence on decision-making and implementation (Brugha & Varvasovszky, 2000). It is widely used in planning and public management to identify whose interests must be taken into account and who has the power to enable or block change (Bryson, 2004; Reed et al., 2009).

2.6 Managerial issues of bus terminals

The basic managerial functions are planning, organizing, staffing, directing and controlling (Smallbusiness, 2019). In this study, the managerial issues of bus terminals refer to how these functions are performed by the institutions involved in terminal operation, and to the relationships among those institutions.

2.7 Institutional framework for terminal management

Successful provision of a terminal requires a formal set of organizational structures, rules and norms (SGArchitects, 2015). A stable institutional framework must specify who will operate the terminal and how, how the facility will be maintained, how revenue for its development and maintenance will be generated, and what business model, if any, will apply (Finn & Mulley, 2011). Several agencies and organizations may combine in different arrangements to develop, maintain and operate a terminal (SGArchitects, 2015):

1**. Bus service providers**: government departments and municipalities, public corporations and private companies.

2. Regulatory and enforcement bodies: bodies, often local authorities, that regulate terminal operations and services and establish the controls necessary for managing the terminal facility.

3. The private sector: firms that finance and maintain the terminal facility, often in partnership with the public sector.

4. Local authorities: bodies that supervise the development, maintenance and operation of the terminal, both within its boundary and in the surrounding area.

In an ideal institutional framework, the agencies involved maintain well-defined, cooperative relationships and operate according to clearly defined roles set out in legally binding contracts (Finn & Mulley, 2011). In practice, bus sectors in developing countries often depart from this ideal: regulation tends to swing between public monopoly and fragmented, weakly regulated private provision, and the institutions needed to plan and enforce service standards are frequently missing (Gwilliam, 2008; Finn & Mulley, 2011). Salazar Ferro, Behrens and Wilkinson (2013) describe many such systems as hybrid: formally regulated services and infrastructure coexist with informal operators and practices, and the boundary between the two is negotiated rather than fixed. Dhaka's terminals, which are publicly owned but, as later chapters show, informally controlled, can be read as hybrid systems of this kind.

2.8 Urban informality as a mode of governance

Informality is commonly understood as activity that falls outside state regulation. Planning scholarship, however, increasingly treats it as a mode of urban governance in its own right—an organizing logic through which the state itself allocates land, services and opportunities (Roy, 2005). In South Asian cities, Roy (2009) argues, informality is produced by the state itself, through its discretionary power to determine what is permitted, tolerated or excluded. According to Hackenbroch and Hossain (2012), informality is an important mode of urban governance and an expression of particular constellations of power: a mode of state regulation practiced with careful regard to the power and influence of the groups concerned.

The state tolerates the informal activities of the powerful, and even supports them by extending infrastructure and services, while criminalizing similar activities by groups less able to exert power. Over time, the practice and stabilization of such institutional forms widen the gap between the privileged and those unable to participate on the same terms, producing an uneven spatiality of power (Yiftachel, 2009).

Informal practices therefore tend to produce exclusionary and discriminatory outcomes. Hackenbroch and Hossain (2012) describe the resulting informal negotiations as the 'organized encroachment of the powerful' (see also Roy & AlSayyad, 2004). The phrase deliberately inverts Bayat's (2000) notion of the 'quiet encroachment of the ordinary', the gradual advance of the urban poor onto public space and services: in Dhaka, Hackenbroch and Hossain argue, it is the powerful rather than the poor who encroach, and they do so with the support of the state. This inversion is central to the present study, because the actors who control Dhaka's bus terminals are not marginal groups but organized and politically connected ones.

In Dhaka, where statutory institutions are fragile, access to livelihood assets in low-income settlements is negotiated and contested within hybrid institutional settings and uneven power

structures, through continuous negotiation and shifting relationships between statutory and non-statutory actors (Hackenbroch & Hossain, 2012). Public space in informal settlements is contested and tends to fall under the control of powerful individuals and institutions backed by the party system. Informal negotiation has become a regular, everyday means of gaining access to urban amenities and services in Dhaka (Hackenbroch, 2010).

According to Hackenbroch and Hossain (2012), local leaders run associations that facilitate communication and negotiation with political leaders and government officials at various levels. These relationships rest on mutual support: local leaders help to organize party meetings and election campaigns and, in return, receive protection and legitimation for informal businesses and local regulations from higher-level leaders. They also maintain relationships with government staff, including the police, through regular informal payments. Actors who acquire power through party politics can dominate negotiation and decision-making, whereas those without political affiliation, or affiliated with the opposition, have very limited voice. Political scientists describe such durable, unwritten rules operating alongside formal ones as informal institutions, which may complement, accommodate, compete with or substitute for the formal rules they shadow (Helmke & Levitsky, 2004).

Most of this literature concerns housing, land and basic services. Banks, Lombard and Mitlin (2020) argue, however, that informality is better understood as an analytical lens on the relationship between state and non-state actors across sectors than as a property of particular places or groups. Applying this lens to transport infrastructure is the step this study takes. A bus terminal is, like a settlement or a market, a public space whose use is governed by negotiated arrangements among statutory authorities, organized interest groups and political patrons, and the concepts reviewed in this section—selective tolerance by the state, organized encroachment by the powerful and informal institutions that shadow formal rules—are used in Chapters 7 to 9 to interpret how Dhaka's terminals are managed.

2.9 Informality in the management of bus transport

A range of rent-seeking practices affects all aspects of bus operation and regulation in Dhaka and makes new investment unprofitable. To stay in business, bus owners must pay extensive formal and informal fees, bribes and 'speed money', and the transport authorities lack both the capacity to monitor these practices and the political and institutional support to curb them (Karmakar & Alam, 2016). Bus associations, meanwhile, are allied with political leaders who are in turn backed by transport businesses (Sen, 2016).

Akbar and Campos (2009) observe that bus terminals in Dhaka have been turned over to politically connected operating companies. Control of a terminal gives operators rent-seeking

opportunities, including proceeds from illicit trade and the shipment of contraband goods. The arrangement persists because it serves the interests of powerful stakeholders: the capacity of regulatory institutions is kept weak, the City Corporation lacks the staff to take action, and coordination within the City Corporation, and between BRTA and other agencies, is poor.

Similar dynamics have been documented elsewhere in the Global South. Informal and semi-formal operators, organized through owners' associations and route cartels, provide much of the public transport in cities of Africa, Asia and Latin America (Cervero & Golub, 2007; Behrens et al., 2016), and studies of Nairobi and Dar es Salaam show how transport associations, labor organizations and political patrons shape the outcomes of public transport policy (Klopp, 2012; Rizzo, 2017). Closest to the present study, Agbiboa (2018) shows how the motor parks of Lagos are governed by touts absorbed into the road transport workers' union, who collect levies from operators, enforce the union's authority and mobilize votes for allied politicians; because these arrangements serve both union and political interests, official attempts to remove the touts have largely failed. Agbiboa's work establishes that bus terminals can be sites of informal and predatory governance. What remains less examined is how such governance shapes the spatial and functional organization of the terminal itself—where buses park, where passengers board and alight, and which facilities are used—and how it interacts with the planning standards that terminals are designed to meet. This is the question this study takes up.

2.10 Analytical tools

Venn diagram

A Venn diagram depicts the relationships among the institutions, organizations, groups and important individuals in a community, as well as people's views of their importance. It also indicates how close the relationships between these organizations and groups are (Kumar, 2002).

Timeline: A timeline is a participatory rural appraisal (PRA) method used to investigate the temporal dimension of a subject from a historical perspective. It records events in the order in which they occurred, as recalled by local people, and thus identifies the historical landmarks of a community (NIRD, n.d.).

Cause–effect diagram: A cause–effect diagram examines why something happened, or might happen, by organizing potential causes into smaller categories. It is also useful for showing the relationships between contributing factors (Kumar, 2002).

Stakeholder mapping

Stakeholder mapping plots stakeholders against two variables in order to identify those with the greatest potential impact on an initiative. Commonly used matrices include the power–interest, importance–influence, influence–interest and power–influence matrices (MacArthur, 1997; Reed et al., 2009).

2.11 Research gap and analytical framework

The literature reviewed above tends to treat the problems of bus terminals in one of two ways. Design guidance (Chiara & Crosbie, 1989; SGArchitects, 2015) treats them as problems of provision and layout, to be solved by supplying the right facilities in the right sequence. The governance literature (Akbar & Campos, 2009; Hackenbroch & Hossain, 2012; Klopp, 2012) treats them as problems of power and rent-seeking but rarely examines how these play out in the physical operation of a facility. Existing studies of Dhaka's terminals (Rahman et al., 2008) document deficiencies in facilities but do not explain why facilities that do exist go unused. To address this gap, the study brings together two bodies of literature that are rarely combined. Research on the governance of public transport in the Global South (Cervero & Golub, 2007; Gwilliam, 2008; Salazar Ferro et al., 2013; Agbiboa, 2018) identifies the actors and mechanisms—operator associations, unions, route permits and political patronage—through which bus systems come to be controlled by organized interests rather than by regulators. Research on urban informality as a mode of governance (Roy, 2005, 2009; Hackenbroch & Hossain, 2012; Banks et al., 2020) explains why the state tolerates and draws on such arrangements, and why they persist. The first lens is used to trace how the terminals are actually managed; the second to interpret why formal authority over them is so weak.

This study connects these perspectives. It treats the functional arrangement of a terminal—where buses park, where passengers board and alight, where tickets are sold—as an observable outcome, and the configuration of formal and informal institutions that govern the terminal as a potential explanation of that outcome. The analysis proceeds in three steps. First, the existing functional arrangement of each terminal is compared with the standard arrangement (Chapters 5 and 6). Second, the actors involved in terminal management, and the distribution of formal power and actual influence among them, are mapped (Chapter 7). Third, cause–effect analysis traces how specific managerial failures produce specific functional deficiencies (Chapter 8). The working proposition is that where informal actors exercise greater influence than the formal authority, the spatial organization of the terminal will reflect their interests rather than the requirements of passengers and planning standards.

This chapter describes the research design and the methods used to achieve the objectives of the study. The study adopts a qualitative multiple-case design (Yin, 2018), in which each of Dhaka's three designated inter-district terminals is treated as a case. Data from field observation, key informant interviews, focus group discussions and documentary sources were triangulated to strengthen the credibility of the findings. The overall methodological framework is shown in Figure 3.1.

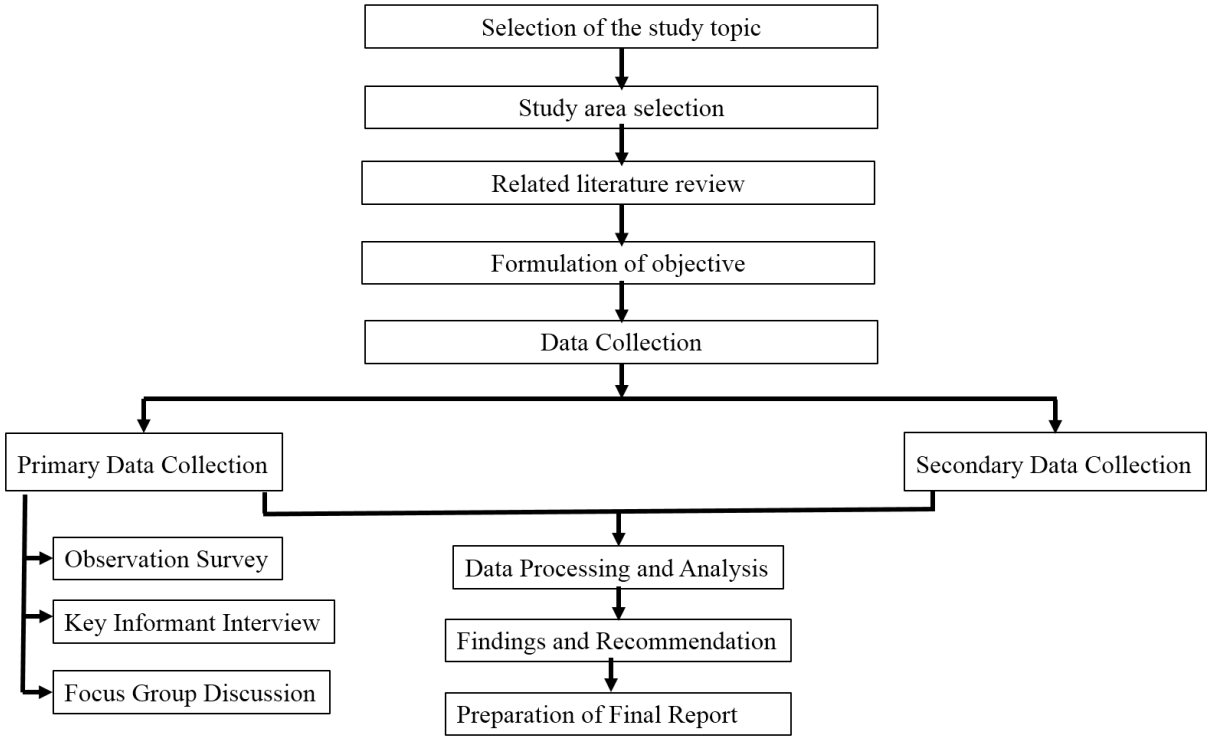


Figure 3.1: Methodological Framework

Table 3.2 links each research question to its data sources, analytical tools and the chapters in which it is answered.

Table 3.2: Research questions, data sources and analytical tools

Research question	Data sources	Analytical tools	Chapters
RQ1. How do the facilities, and the flows of buses and passengers, in each terminal compare	Observation surveys (checklists in Appendices A and B); FGDs with passengers; Bus Terminal	Facility inventory against Table 2.1; mapping of bus and passenger flows; comparison with the	5, 6

Research question	Data sources	Analytical tools	Chapters
with a standard functional arrangement?	Design Guidelines (SGArchitects, 2015)	standard arrangement (Figure 6.1)	
RQ2. Which actors operate and manage the terminals, and how are formal power and actual influence distributed among them?	KIIs with city corporation officials, owners' associations, labor unions and an academic expert; documents and news reports	Stakeholder identification (internal and external); power–influence matrix; Venn diagram	7
RQ3. Through which mechanisms do management arrangements produce the functional deficiencies, and what reforms would address them?	KIIs; FGDs; observation; documents and news reports	Timeline; cause–effect diagrams; cross-terminal comparison; SWOT analysis	8, 9

3.1 Selection of the study topic:

The study set out to identify the problems of Dhaka's bus terminals. Such problems can be examined from several perspectives; this study focuses on two that are closely linked—functional arrangement and management—and on the relationship between them.

3.2 Study area selection:

Inter-district bus terminals are critical infrastructure components of a bus-based transit system. Dhaka has three designated inter-district terminals—Sayedabad, Mohakhali and Gabtoli—located within the administrative boundaries of Dhaka South City Corporation (DSCC) and Dhaka North City Corporation (DNCC) (DTCA, 2016). All three were selected as the study area; Fulbaria terminal was excluded because it is not a designated inter-district terminal. Studying all three designated terminals allows comparison across cases that share the same regulatory framework but differ in size, managing city corporation and operating conditions.

3.3 Related literature review:

Previous studies were collected from theses, journal articles, books and newspaper reports. Few academic studies of Dhaka's bus terminals were found, so newspaper reports on the terminals' problems and management were also reviewed.

3.4 Formulation of objectives

Two specific objectives, and the research questions derived from them (Section 1.2), were formulated to guide the study.

3.5 Data Collection

3.5.1 Primary Data Collection

Primary data were collected using three methods: observation surveys, key informant interviews (KIIs) and focus group discussions (FGDs).

Observation Survey:

Observation surveys were conducted to record the location and condition of terminal facilities and to assess whether the functional arrangement of each terminal conforms to the standard arrangement. The location and condition of bus parking, parking for other vehicles, passenger loading and unloading areas, toilets, bus washing and cleaning facilities, passenger sheds and waiting rooms were recorded using the checklists in Appendices A and B. The surveys were conducted between November 2018 and February 2019.

Key Informant Interview

Key informants were selected purposively to represent the internal and external stakeholders of the three terminals, as defined in Section 2.5. Ten interviews were conducted with city corporation officials, representatives of bus owners' associations and labor unions, and an academic transport expert (Table 3.1).

Table 3.1: Directory of Key informant interviewee

Sl. no.	Name	Designation	Date of Interview
1	Abdul Lotif Khan	General Manager (Transport), Dhaka North City Corporation (Gabtoli and Mohakhali Bus Terminal)	26 Nov, 2018
2	Joshoi Ali	Assistant Manager, Dhaka North City Corporation (Gabtoli Bus Terminal, Mohakhali Bus Terminal)	20 Nov, 2018
3	Golam Morshed	Assistant Manager, Dhaka South City Corporation (Sayedabad Bus Terminal)	4 Dec, 2018
4	Ariful Islam	Supervisor , Dhaka South City Corporation (Sayedabad Bus Terminal)	26 Nov, 2018

5	"A",	Bangladesh bus truck owner association,(Gabtoli Bus Terminal)	27 Nov, 2018
6	"B"	Bangladesh Sarak Poribohon Malik Somiti,(Sayedabad and Mohakhali Bus Terminal)	5 Dec, 2018
7	"C"	Dhaka Zilla Janbahon Sromik Union (Reg No.974), (Gabtoli Bus Terminal)	26 Nov, 2018
8	"D"	Dhaka Mohanogor Sorok Poribohon Sromik Union (Reg No.-2195), (Sayedabad Bus terminal)	4 Dec, 2018
9	"E"	Dhaka Zilla Bus Minibus Sarak Poribohon Sromik Union (Reg No.2415), Mohakhali Bus Terminal	5 Dec, 2018
10	Dr.Shamsul Haque,	Professor, Department of Civil Engineering, BUET	4 Feb, 2019
Note: Letters are used in place of names to protect informants' anonymity.			

The interviews explored why the terminal system does not work effectively and examined the managerial issues of the terminals (Appendices C and E). They also served to cross-check the observation data from the perspective of operators and managers. Because informants discussed sensitive matters, including informal payments and political influence, the names and designations of owners' association and labor union representatives have been withheld to protect their anonymity and confidentiality. For the same reason, the audio recorder was not shown to informants and no photographs were taken during the interviews. The interviews were conducted over four months, from November 2018 to March 2019.

Focus Group Discussion (FGD)

Three focus group discussions were conducted with passengers, one at each terminal, in the last week of January 2019 (Appendices D and J). The FGDs were used to cross-check the observation data against passengers' perceptions; participants were asked about the condition of terminal facilities and why they did or did not use them.

Table 3.3 summarizes the coverage of the primary data by stakeholder group and terminal. Each internal stakeholder group was represented at each terminal, and each terminal was covered by observation and a passenger FGD.

Table 3.3: Coverage of primary data by stakeholder group and terminal

Stakeholder group / method	**Sayedabad**	**Gabtoli**	**Mohakhali**
City corporation officials (KII)	KII 3, 4 (DSCC)	KII 1, 2 (DNCC)	KII 1, 2 (DNCC)

Stakeholder group / method	Sayedabad	Gabtoli	Mohakhali
Bus owners' associations (KII)	KII 6	KII 5	KII 6
Labor unions (KII)	KII 8	KII 7	KII 9
Academic expert (KII)	KII 10	KII 10	KII 10
Passengers (FGD)	FGD 01	FGD 02	FGD 03
Observation survey	Nov 2018–Feb 2019	Nov 2018–Feb 2019	Nov 2018–Feb 2019
External agencies (DMP, BRTA, RTC, DTCA, LGRD&C)	Not interviewed	Not interviewed	Not interviewed

Note: KII numbers refer to Table 3.1. KIIs 1, 2, 6 and 10 covered more than one terminal. The roles of external agencies are described from the accounts of internal stakeholders and from secondary sources.

3.5.2 Secondary Data Collection:

The standard functional arrangement of a bus terminal was taken from the Bus Terminal Design Guidelines (SGArchitects, 2015) and compared with the existing arrangements of Dhaka's terminals. Newspaper reports were used as a secondary source on the terminals' problems.

3.6 Data Processing and Analysis:

Maps of the three terminals were prepared from the observation surveys with the aid of the Detailed Area Plan (DAP), 2015. The standard functional arrangement was then compared with the existing arrangement of each terminal, using the observation and FGD data.

The comparison with the standard proceeded in two steps. First, the facilities present at each terminal were recorded against the facility requirements in Table 2.1, using the checklists in Appendices A and B. The results are summarized in those appendices, in which each facility is classified as present, partly present (insufficient or not used for its intended purpose) or absent. Second, the observed flows of buses and passengers were traced and compared with the sequence of functions in the standard arrangement (Figure 6.1) to identify where, and why, the observed flows depart from it.

The interview and focus group material was analyzed thematically. Notes from each interview and discussion were organized under themes derived from the study's objectives: (i) facilities and their use; (ii) bus and passenger flows; (iii) parking and route permits; (iv) ticketing; (v) revenue and toll collection; (vi) staffing and recruitment; (vii) crew welfare; and (viii) relationships among stakeholders. Evidence on each theme was then compared across the three terminals and across types of informant (officials, owners, workers and passengers), and

checked against the observation data where possible. Table 3.4 shows the sources on which each principal finding rests.

Table 3.4: Evidence base for the principal findings

Principal finding	Observation	KII	FGD	Documents	Sections
Passengers board and alight on roads outside the terminals	Yes	Yes	Yes	–	5.1.7, 5.2.6, 5.3.7, 6.2–6.4
Ticket counters operate outside the terminals	Yes	Yes	Yes	Yes	5.1.8, 6.2.3, 8.2.2
Terminals are used as depots; parking demand exceeds supply	Yes	Yes	–	Yes	5.1.2, 5.2.2, 5.3.2, 8.2.2
Passenger amenities are absent or under-used	Yes	Yes	Yes	Yes	5.1–5.3, 6.2.3, 6.3.3
No resting rooms for bus crews	Yes	Yes	–	–	6.2.3–6.4.3, 8.2.2
Owners' associations and unions have more influence than the City Corporation	–	Yes	–	Yes	7.3
Tolls are collected informally by union collectors	Partly	Yes	–	Yes	5.1.3, 8.2.1
Staff shortage and informal daily-wage recruitment	–	Yes	–	–	8.2.1
Route permits are obtained through owners' associations without depot requirements	–	Yes	–	–	8.2.2
Political involvement in terminal management varies with national political change	–	Yes	–	Yes	8.1, 9

Note: 'Partly' indicates indirect evidence (e.g., the fee paid by CNG drivers to the labor union observed at Sayedabad). Documentary sources are Rahman et al. (2008), Akbar and Campos (2009), Karmakar and Alam (2016), Sen (2016) and the news reports cited in Chapter 9. Findings on functional arrangement rest on three or more sources; findings on revenue, staffing, route permits and political influence rest mainly on interview accounts and documents and could not be verified against official records (Section 1.4).

Several tools were used to identify the stakeholders and analyze the managerial issues of the terminals, as described below.

3.6.1 Power Influence matrix:

A power–influence matrix places each institution according to its level of formal authority (power) and its level of actual involvement in decision-making (influence). The matrix was constructed from the interviews with labor union representatives and city corporation officials, and the placement of each stakeholder reflects the authors' interpretation of these interviews.

3.6.2 Venn diagram:

The Venn diagram was prepared from the interviews with labor union and owners' association representatives and city corporation officials to represent institutional relationships. Circles of different sizes represent institutions or individuals, with larger circles indicating greater influence. The direction of the arrows shows the direction of interaction among city corporation officials, labor unions, owners' associations and outside organizations, and their width shows the intensity of interaction between two stakeholders.

3.6.3 Timeline:

A timeline of the three terminals was prepared through an interactive discussion with a city corporation official who had worked at the terminals for more than 20 years. The discussion focused on the evolution of the terminals' management system.

3.6.4 Cause-effect diagram:

Managerial problems were identified from the KIIs, and the causes and effects of the principal problems were analyzed using cause–effect diagrams. Causes and effects were first listed, and sub-causes and sub-effects were then identified where informants described them.

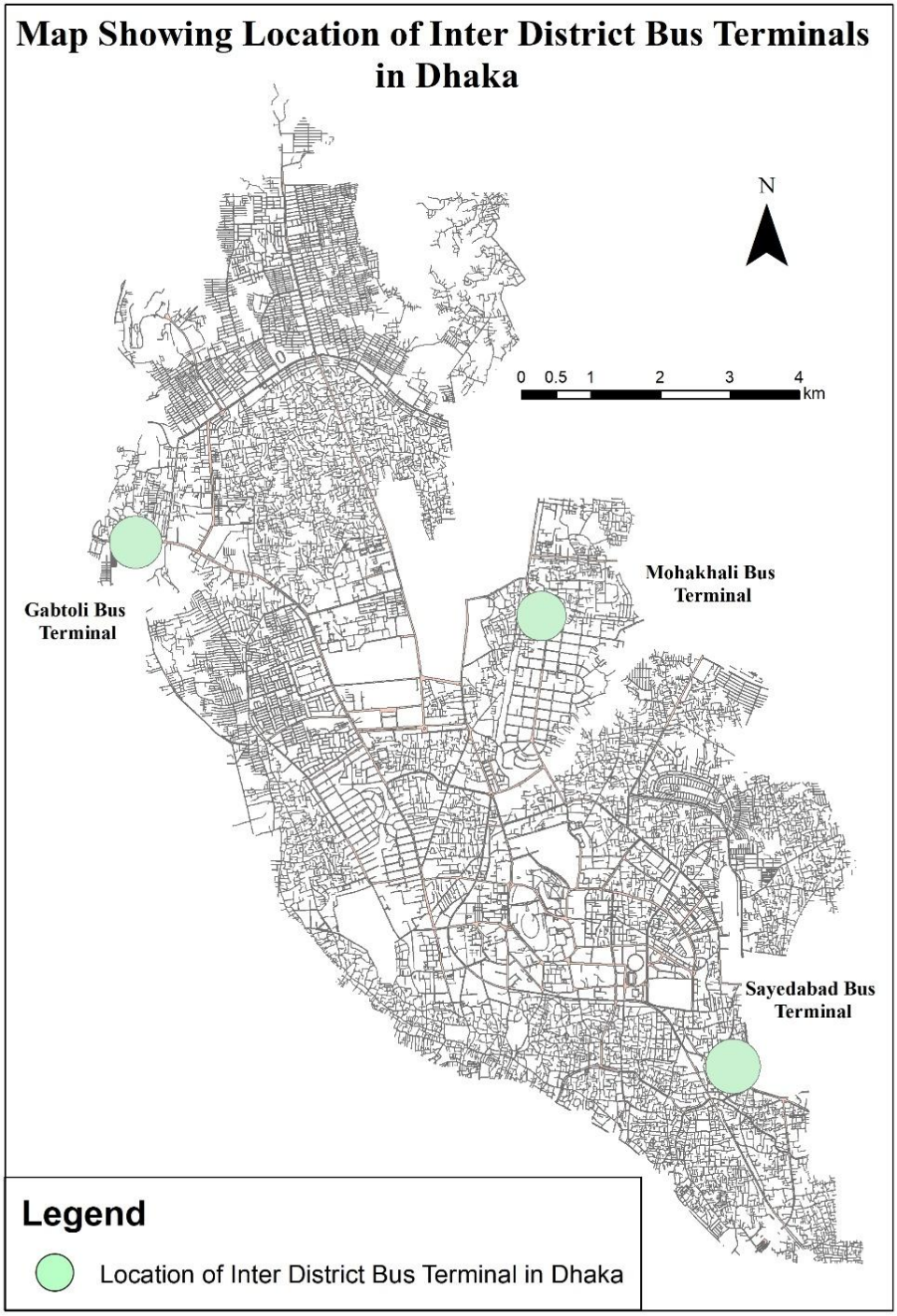
3.7 Findings and Recommendation

Major findings were derived from the analysis of the information collected, and recommendations were formulated to address the managerial and functional problems identified.

3.8 Preparation of Final Report

The final report was prepared by compiling and synthesizing the information collected and analyzed.

The study area comprises the three designated inter-district bus terminals of Dhaka: Sayedabad, Mohakhali and Gabtoli (Map 4.1).



Map 4.1: Map of Study Area

Source: Prepared by authors

Sayedabad Bus terminal

Sayedabad bus terminal is located in the eastern part of Dhaka and covers an area of 40,500 m². About 2,000 buses operate from the terminal each day on 87 routes, serving four zones: the eastern, north-eastern, south-eastern and southern districts of Bangladesh. Buses using the terminal travel on three national highways:



Figure 4.1: Sayedabad Bus terminal

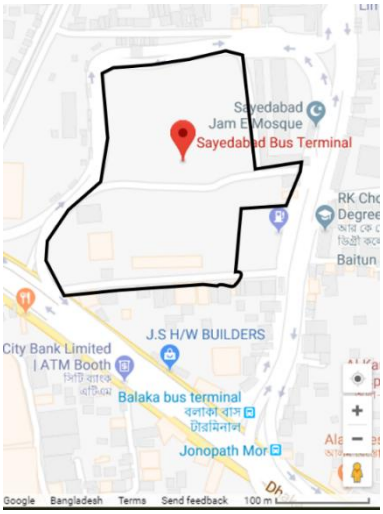


Figure 4.2: Boundary of Sayedabad Bus terminal (Google map, 2019)

- N1: Dhaka (Jatrabari)–Comilla (Mainamati)–Chittagong–Teknaf Road
- N8: Dhaka (Jatrabari)–Mawa–Bhanga–Barisal–Patuakhali Road
- N2: Dhaka (Katchpur)–Bhairab–Jagadishpur–Shaistaganj–Sylhet–Tamabil–Jaflong Road

Gabtoli Bus terminal

Gabtoli bus terminal, established in 1945, is located in the north-western part of the city, adjacent to the Dhaka–Aricha highway. Its buses mainly serve the northern and south-western districts of Bangladesh. The terminal covers 123,400 m², and about 2,200 inter-district buses operate from it on 61 routes, mainly using the N5 highway (Dhaka (Mirpur)–Utholi–Paturia–Natakhola–Kashinathpur–Bogra–Rangpur).



Figure 4.3: Gabtoli Bus terminal

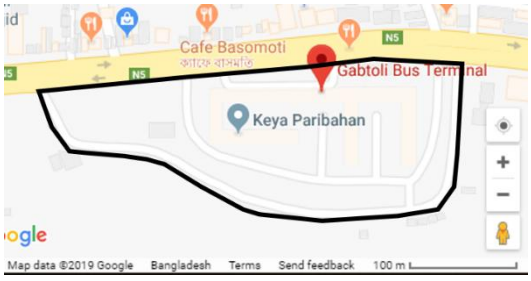


Figure 4.4: Boundary of Gabtoli bus terminal

(Google map, 2019)

Mohakhali Bus terminal

Mohakhali inter-district bus terminal was built in 1984 and is located in the northern part of

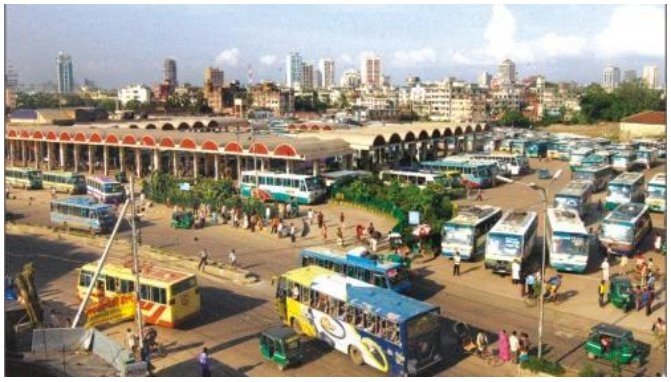


Figure 6: Mohakhali Bus terminal

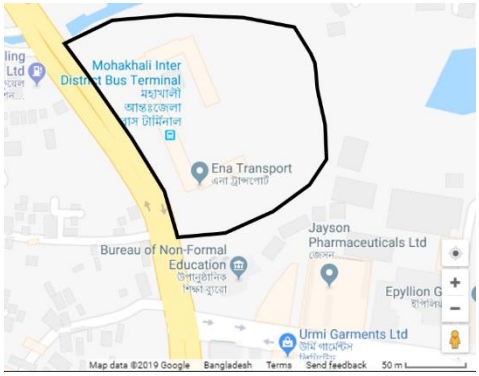


Figure 4.6: Boundary of Mohakhali bus terminal

(Google map, 2019)

Dhaka, near the Mohakhali intersection and adjacent to the Tejgaon industrial area. The terminal covers 36,400 m², and around 800 inter-district buses operate from it on 60 routes, mainly using the N3 highway (Dhaka (Banani)–Joydebpur–Mymensingh Road).

Bus terminals provide a range of facilities and services to support their functional system. This chapter describes the existing condition of the facilities and services at each of the three terminals.

5.1 Existing terminal facilities and services of Sayedabad Bus Terminal:

Sayedabad terminal provides a range of facilities for passengers and staff. The existing facilities are described below and mapped in Map 5.1.

5.1.1 Entry and exit point

The entry point of Sayedabad terminal, 30 feet wide, is located on the eastern side near the police control room (Figure 5.1). The two exit points, 28 and 29.5 feet wide, are separated by the terminal building (Rahman et al., 2008) and are located in the south-western part of the terminal near the Dhaka–Sylhet highway (Figure 5.2). The Mayor Hanif flyover has one entry ramp and two exit ramps inside the terminal (Appendix G).



Figure 5.1: Entry point of Sayedabad bus



Figure 5.2: Exit point of Sayedabad bus

Source: Field Survey, 2018

5.1.2 Parking spaces for terminal buses

The terminal can accommodate 500 buses at a time (Figure 5.3). Because the number of buses using the terminal exceeds the parking supply, the designated spaces are not used as intended: buses park haphazardly, newly arriving buses have difficulty reaching designated spaces (Appendix G), and congestion builds up inside the terminal.

5.1.3 Parking spaces for other transports

Parking for other vehicles is provided near the terminal entrance, mainly for rickshaws and CNG auto-rickshaws. Rickshaws park free of charge, whereas CNG auto-rickshaw drivers pay Tk 20 per trip to the labor union—an early indication of the informal revenue collection examined in Chapter 8.



Figure 5.3: Parking facility of Sayedabad bus terminal



Figure 5.4: Washing facility of Sayedabad bus terminal

Source: Field Survey, 2018

5.1.4 Bus washing and cleaning Facilities

Only one bus washing ramp exists, which is insufficient for a terminal of this size. Buses are therefore washed in the parking area, which leaves it dirty (Figure 5.4).

5.1.5 Repair facilities

Small repair shops are scattered throughout the terminal (Figure 5.5), and buses are repaired across the whole premises, even though servicing and maintenance are not permitted in the



Figure 5.5: Repairing facility of Sayedabad bus terminal



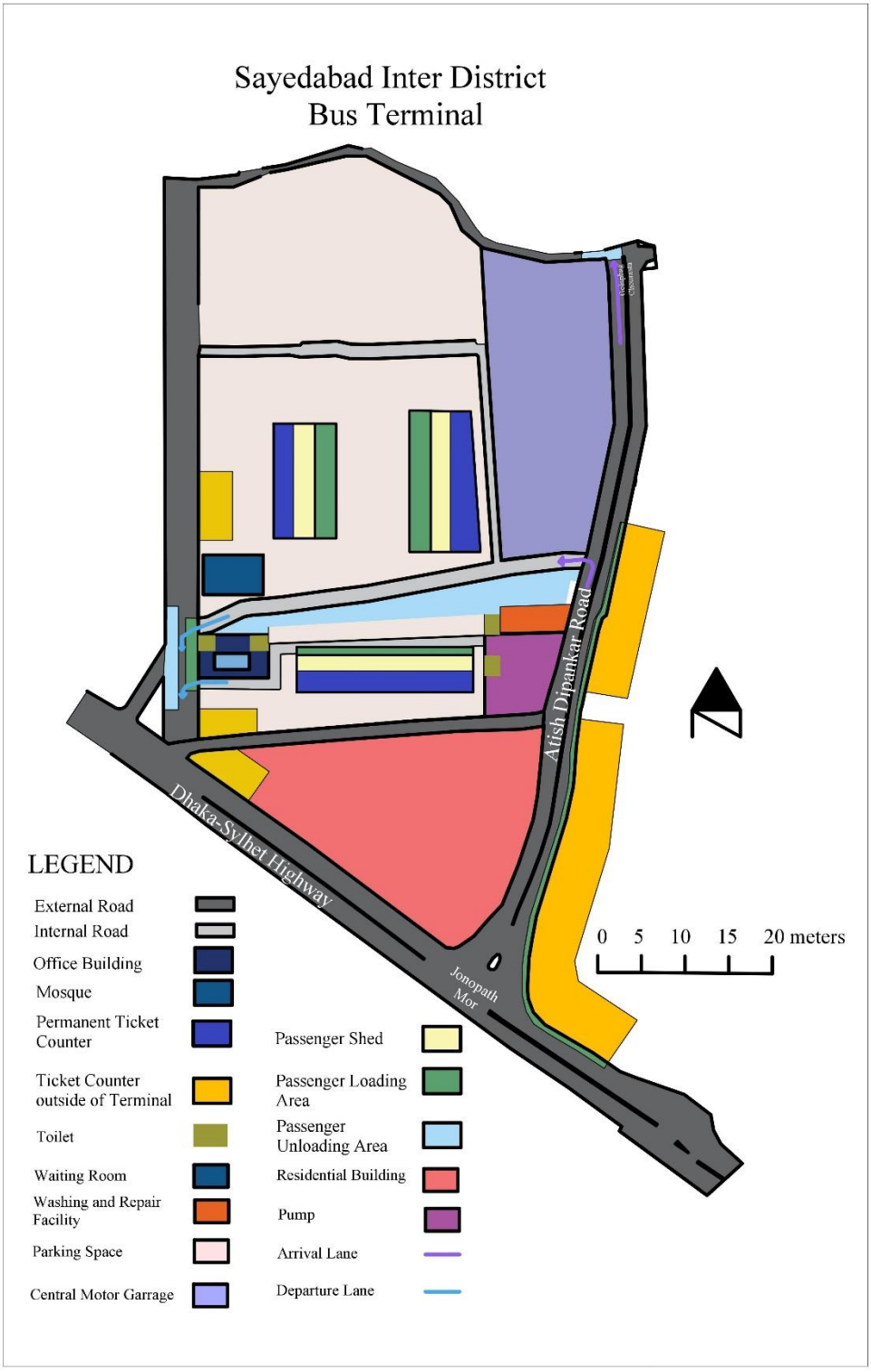
Figure 5.6: Waiting room of Sayedabad bus terminal

Source: Field Survey, 2018

terminal and should be carried out in operators' own workshops. A petrol pump inside the terminal provides refueling.

5.1.6 Arrival lanes and departure lanes

The terminal has one designated arrival lane and three designated departure lanes. The Sayedabad bus owners' association sets the time allocation for arriving and departing buses without consulting the City Corporation and without regard to actual demand for the lanes. Designated bus bays are provided in front of the passenger



Map 5.1: Map of Sayedabad Bus Terminal

Source: Prepared by authors

platform. Although the bays are intended for boarding and alighting, they are not used for that purpose; buses generally pick up and set down passengers outside the terminal.

5.1.7 Passenger loading and offloading area

Most passengers board at Jonopod Mor on the Dhaka–Sylhet highway (Appendix G); only a few board inside the terminal. Passengers alight near the entrance of the terminal building and at Jonopod Mor.

5.1.8 Ticket counter

Ticket counters are grouped into three zones serving the north-eastern, south-eastern and southern districts (Figure 5.8), with temporary counters for nearby districts such as Comilla, Narayanganj and Chandpur. Few passengers buy tickets at these counters, because the same bus companies also operate counters outside the terminal, where most passengers buy their tickets (Appendix G). The companies compete intensely with one another for passengers. The counters inside the terminal average 30–40 sq ft in area.

5.1.9 Waiting room:

The terminal building has two storeys. The passenger waiting room is on the ground floor (Figure 5.6), and the offices of the City Corporation and of the workers' and owners' associations are on the first floor. There are 36 shops and canteens in the terminal, and hawkers operate around the terminal building.

5.1.10 Toilet facility

Sayedabad terminal has four public toilets. Two, located inside the terminal building, are maintained jointly by the City Corporation and WaterAid and are separated for women and



Figure 5.7: Toilet Facility of Sayedabad bus terminal



Figure 5.8: Ticket counter of Sayedabad

Source: Field Survey, 2018

men (Figure 5.7). The other two, near the terminal entrance, are maintained by the City Corporation alone and are shared by men and women.

5.1.11 Passenger Shed

Passenger sheds are located in front of the ticket counters, but few people use them because most passengers buy tickets from counters outside the terminal (Appendix G).

5.1.12 Information desk facility:

There is currently no information desk. One existed previously but was little used. Nor is there any signage informing passengers of arrival and departure times or of the location of facilities such as the waiting room and toilets. As a result, passengers have difficulty finding facilities and their buses.

5.2 Existing terminal facilities and services of Gabtoli Bus Terminal

Terminal facilities can be divided into passenger amenities, amenities for terminal and bus staff, and general terminal facilities. Not all of these are available at Gabtoli terminal. Some amenities are in good condition, but most passengers do not use them. The existing facilities, identified through the observation survey, are described below and mapped in Map 5.2.

5.2.1 Entry and Exit

The terminal has two entry points and two exit points. One entry gate, located in front of the terminal building, is used by pedestrians and feeder services (Figure 5.9). The other



Figure 5.9: Entry point of Gabtoli bus



Figure 5.10: Exit point of Gabtoli bus

Source: Field

entry gate is located on the western side and is used by buses entering for servicing and repair. A service road behind the terminal, outside its boundary, is used by city buses. Inter-district buses enter the terminal complex through the main entry on the eastern side, park inside, and depart through the main exit (Figure 5.10). Long-distance buses mainly use the main exit, whereas short-distance inter-district buses use the western entry and, most of the time, park on the highway to pick up passengers.

5.2.2 Parking facility for buses

There is no designated bus parking area. All space not occupied by the terminal building and toilets is used for parking inter-district buses (Figure 5.11). Open spaces outside the terminal boundary are also used for parking, mainly by local buses but also by inter-district buses. Parking space within the terminal is currently insufficient. It is divided into zones for long- and short-distance buses, but haphazard and irregular parking leaves part of it underused, and the overall condition of the parking facility is poor.

5.2.3 Parking facility for other transports

There is no designated parking for other vehicles or feeder services such as CNG auto-rickshaws and taxis, which park on the Dhaka–Aricha highway near the terminal's entry and exit points. Local buses stop on the opposite side of the highway, making it difficult for passengers and visitors to reach them from the terminal.



Figure 5.11: Parking facility of Gabtoli bus

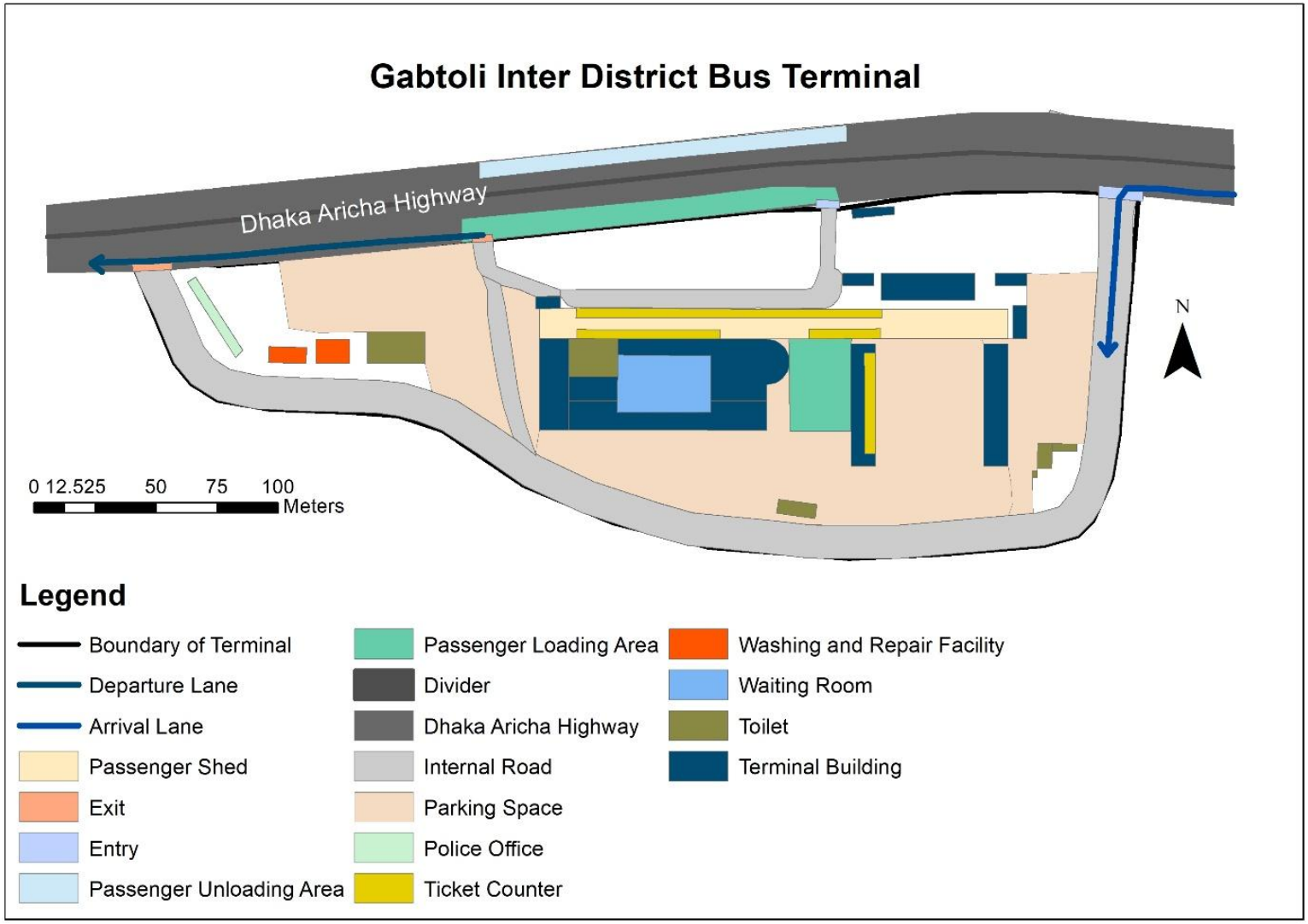


Figure 5.12: Washing facility of Gabtoli bus

Source: Field Survey, 2019

5.2.4 Bus washing and repair facility

A designated bus washing area exists, but it is too small for the number of buses (Appendix H). Staff usually carry water from the washing area and wash buses in the parking area (Figure 5.12). Because there is no designated repair area, bus crews also carry out repairs in the parking area.



Map 5.2: Map of Gabtoli Bus Terminal

Source: Prepared by authors

5.2.5 Arrival and departure lanes

The terminal has one designated arrival lane, at the entry point, and one departure lane, at the exit point. The time allocated for the arrival and departure of buses is 30 to 60 minutes.

5.2.6 Passenger loading and offloading area

There is no designated passenger loading and unloading area. In the absence of bus bays, some passengers board in the parking area and others on the highway (Figure 5.13). Almost all passengers alight outside the terminal, on the opposite side of the highway (Appendix H).

5.2.7 Waiting room

A separate room in the terminal building serves as the passenger waiting room (Figure 5.15), adjoining the office of DNCC. The waiting room is in good condition. About 80 shops and canteens are located around the building (Appendix H).

5.2.8 Ticket counter

More than 50 ticket counters, both permanent and temporary (the latter built of corrugated tin), are located in the passenger shed (Figure 5.14). Some permanent counters are unused (Appendix H). The counters are divided into two zones: those to the right of the entry gate sell tickets for buses to the south of the country, and those to the left for the north. Counter space is allotted by the City Corporation.



Figure 5.13: Passenger loading area of Gabtoli bus terminal

Source: Field



Figure 5.14:

5.2.9 Passenger shed

Passenger sheds are located in front of the ticket counters, where passengers usually sit after buying tickets while waiting for their buses. Before national holidays, the sheds fill up and cannot accommodate all passengers (Rahman et al., 2008); most passengers have to stand, and older people and children find the sheds uncomfortable.



Figure 5.15: Waiting room of Gabtoli bus terminal



Figure 5.16: Toilet Facility of Gabtoli bus

Source: Field

5.2.10 Toilet facilities

There are four public toilet facilities at three locations in the terminal (Figure 5.16); each location has nine toilets, with separate facilities for men and women. Users pay Tk 5 per use. The toilets in the waiting room are maintained by WaterAid and are in better condition than the others, and passengers mainly use these.

5.3 Existing terminal facilities and services of Mohakhali Bus Terminal

Mohakhali terminal provides facilities for passengers, bus crews and terminal staff. The existing facilities and their condition are described below and mapped in Map 5.3.

5.3.1 Entry and exit

The terminal has a single entry, divided into two parts by a median (Figure 5.17). Buses use one side, which is 42 feet wide (Rahman et al., 2008); the other side is used by CNG auto-rickshaws and private cars as a pick-up and drop-off point. The exit, used by all buses, is on the south-eastern side (Figure 5.18).

5.3.2 Parking facility for terminal buses:

The terminal has 128 designated parking spaces (Figure 5.19). Demand far exceeds this supply, since on average about 800 buses originate trips from the terminal each day (Rahman et al., 2008), so buses park haphazardly across the premises.



Figure 5.17:



Figure 5.18:

Source: Field

5.3.3 Parking facilities for other transports

A separate space is allotted for paratransit parking (Appendix I). It can accommodate about 80 CNG auto-rickshaws but has no space for rickshaws (Appendix E), and private cars rarely park there.

5.3.4 Bus Washing and Cleaning Facilities

There is only one washing ramp, on the eastern side, which is insufficient for all the buses in the terminal. Because a fee of Tk 40 is charged for using the ramp, bus crews prefer not to use it, and washing takes place across the whole terminal.



Figure 5.19: Parking Facility of Mohakhali bus terminal



Figure 5.20: Waiting Room of Mohakhali bus terminal

Source: Field

5.3.5 Repair facilities

Although a designated repair area has been provided (Figure 5.23), repairs are carried out throughout the terminal (Appendix I).



Figure 5.21:



Figure 5.22: Ticket counter of Mohakhali bus terminal

Source: Field

5.3.6 Arrival and departure lanes

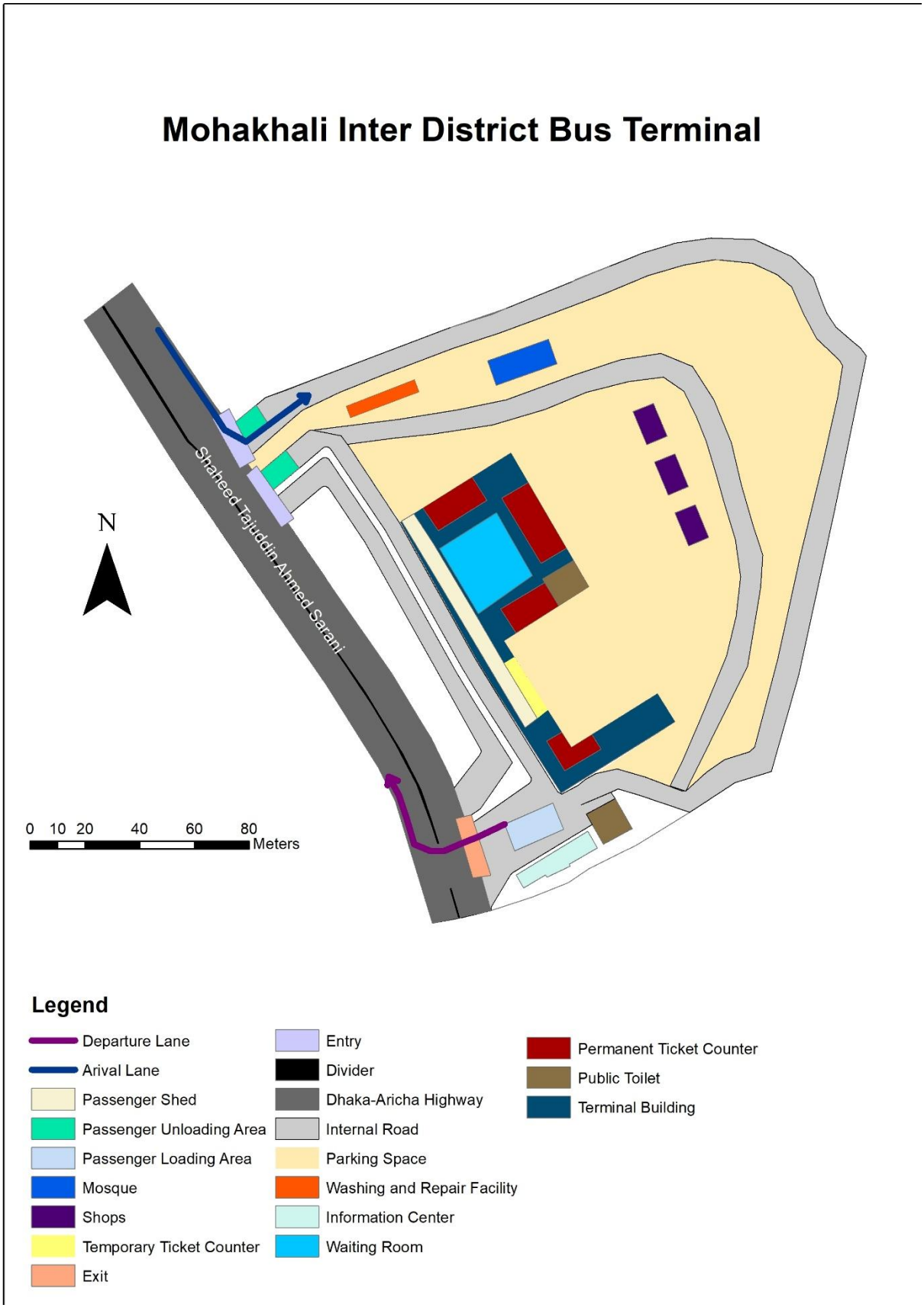
There is one designated arrival lane but no departure lane. The arrival lane is located before the entry point. To depart, buses must make a U-turn on Tajuddin Ahmed Road, creating long queues on the road. There is not enough space inside or outside the terminal for a designated departure lane. Arrival and departure typically take 15 to 60 minutes (Appendix E).

5.3.7 Passenger loading and offloading area

A separate lane was originally provided as a drop-off area for intercity buses, but it is now used by one company (Ena) for parking (Appendix I). Other buses therefore pick up and set down passengers at the exit point. Passengers alight on Shaheed Tajuddin Ahmed Avenue, and buses enter the terminal empty.

5.3.8 Waiting room

The terminal building has two storeys. The waiting room is on the ground floor (Figure 5.20), and the offices of the City Corporation and of the workers' and owners' associations are on the first floor. Seating is inadequate at peak times, and passengers often have to wait standing for long periods. Thirty shops and canteens are located around the building.



Map 5.3: Map of Mohakhali Bus Terminal

Source: Prepared by authors

5.3.9 Ticket counter

There are about 20 designated ticket counters in the terminal building (Figure 5.22). A number of undesignated counters have also been set up alongside them by companies that have not received an allotment (Appendix E). Passenger sheds are located in front of the waiting room.



Figure 5.23: Repair facility of Mohakhali bus terminal



Figure 5.24: Information desk facility of Mohakhali bus terminal

Source: Field

5.3.10 Toilet facility:

Separate toilets for men and women are provided on the ground floor of the terminal building and are maintained by the City Corporation alone (Figure 5.21). A public toilet on the eastern side of the terminal is used mainly by bus crews, and officials have separate toilets on the first floor.

5.3.11 Information desk facility

An information center is located near the exit (Figure 5.24). It provides information on the location of ticket counters and on arrival and departure times.

This chapter compares the functional arrangement of Sayedabad, Gabtoli and Mohakhali terminals with a standard arrangement and identifies the deficiencies of each.

6.1 Standard functional arrangement of a bus terminal:

In planning a bus terminal's circulation and site layout, terminal functions are arranged according to their operational relationships. Figure 6.1 shows the ideal arrangement of terminal functions (SGArchitects, 2015). It sets out the relationships between functions so that the vehicular and pedestrian circulation associated with different activities in the

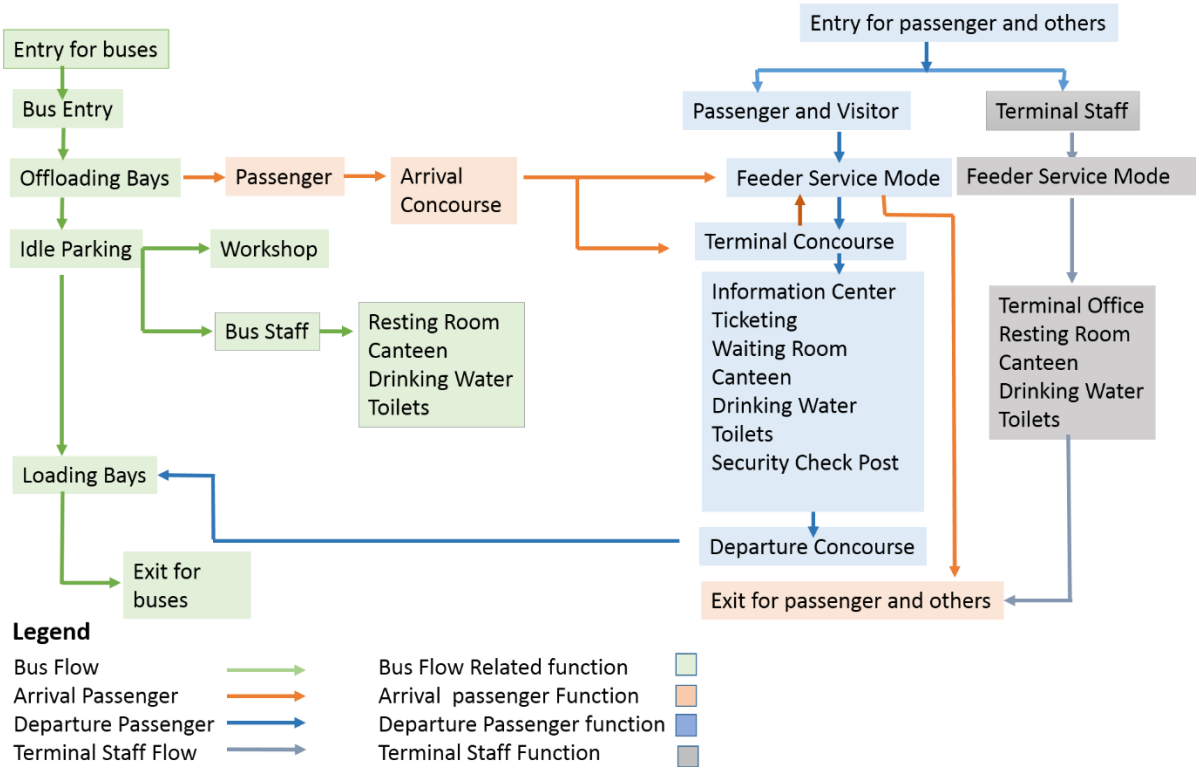


Figure 6.1: Standard Functional Arrangement of a Bus terminal

terminal can be planned without conflict. Buses and passengers have separate entry and exit points. Buses enter through Entry 1 and stop at the offloading bay, a designated place where passengers alight. If a bus has another trip within a few hours, the driver parks it in a designated space in the terminal; otherwise, it returns to the operator's own depot. Bus crews use dedicated facilities in the terminal, such as a dormitory, resting room, canteen, drinking water and toilets. Arriving passengers walk through the arrival concourse to the feeder service zone and leave through a separate exit (Exit 2), or first rest in the terminal concourse. Departing passengers enter through a separate entry (Entry 2), use facilities such as the information center, ticketing,

waiting room, canteen, drinking water, toilets and security check post, and then walk through the departure concourse to the loading bay, where they board buses that leave through Exit 1. Terminal staff also enter through Entry 2, use the terminal office and staff facilities, and leave through Exit 2. The underlying principle is the separation of bus, passenger and staff flows, with each facility placed on the path of the users it serves.

6.2 Functional arrangement of Sayedabad Bus Terminal

Efficient flows of buses and passengers are necessary for a terminal to function smoothly. The functional arrangement of Sayedabad terminal and its deficiencies, based on observation and FGD 01, are described below.

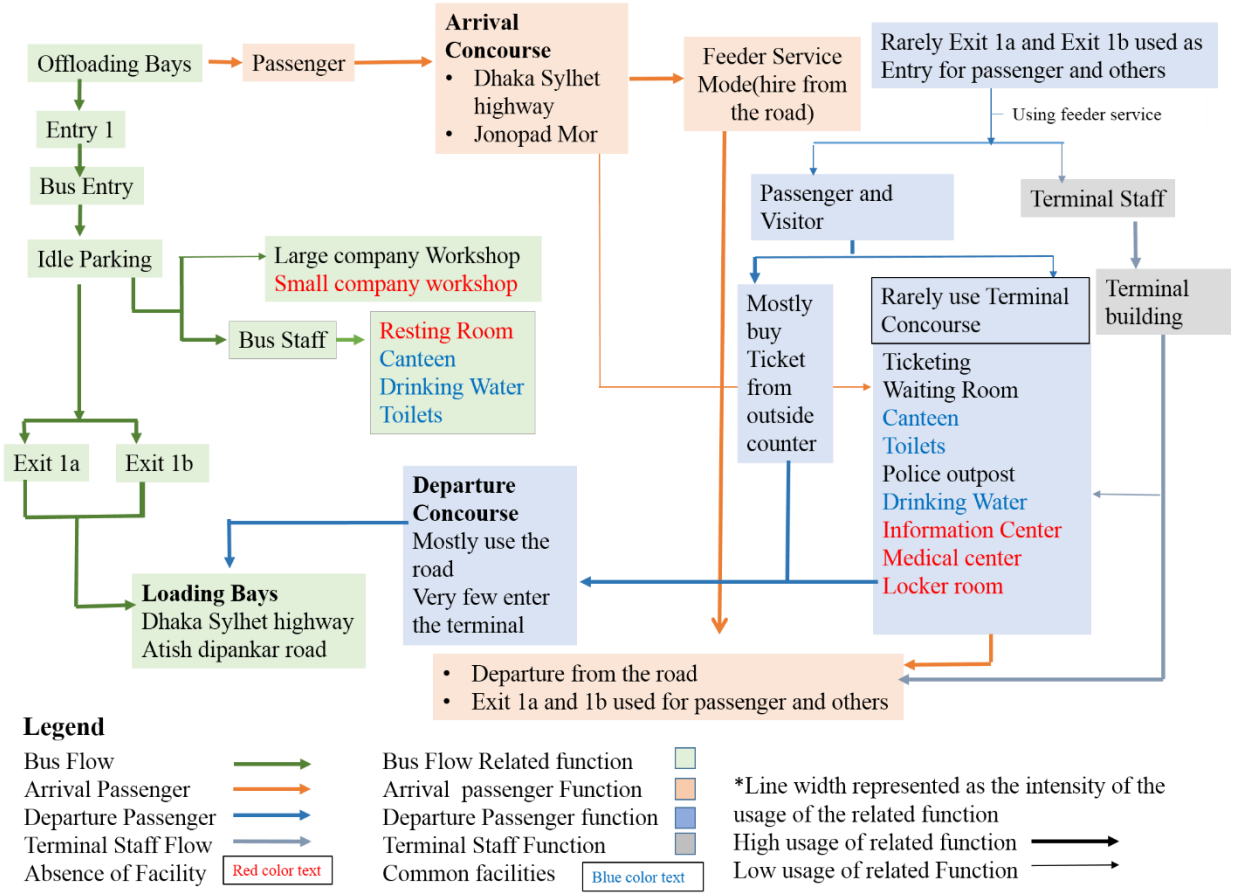


Figure 6.2: Functional Arrangement of Sayedabad Bus

6.2.1 Flow of arrival buses and passengers

- Buses reach the terminal entry mainly via the Dhaka–Sylhet highway; some use the exit ramp of the Hanif flyover.

- Passengers are usually set down at Jonopod Mor, at the flyover ramp inside the terminal and in front of the terminal building (Appendix E).
- Arriving passengers continue their journeys by CNG auto-rickshaw, rickshaw or local bus.
- Buses then enter the terminal empty and use the internal road to reach the designated parking spaces.

6.2.2 Flow of departure buses and passengers

- When it is time for the next trip, buses use the departure lane and leave through either exit (Exit 1a or Exit 1b).
- Some passengers buy tickets at counters inside the terminal and board there, but most buy tickets outside the terminal and board on the Dhaka–Sylhet highway or Atish Dipankar Road. Buses then use the Dhaka–Sylhet highway to reach their destinations.
- A police outpost adjacent to the terminal provides security for passengers; the police sometimes stop and search buses for illegal goods carried by passengers.
- Very few passengers use terminal facilities such as the waiting room and toilets.

FGD-01: Opinion Regarding Functional Arrangement of Sayedabad Bus Terminal Given by Passengers

FGD conducted in the waiting room of Sayedabad Bus Terminal reveals that they use feeder services like CNG, Bus to arrive in the terminal. Since most of the ticket counters are at the outside of the terminal, they are normally unloaded from their vehicles in front of the ticket counters at the outside of the terminal. That's why, they normally don't use the waiting room, toilets provided in the terminal building as well as ticket counters inside the terminal. Some passengers use the toilet facility and must pay 5tk per use which is more than reasonable for them. For the short distance buses, they don't need to buy ticket to get in bus. They are normally offloaded from buses at the outside of the terminal and use the feeder services which are available near the terminal. Because of absence of information center and direction sign in the terminal, they don't get proper guidance which cause waste of their valuable time.

6.2.3 Deficiencies of functional arrangement of Sayedabad bus terminal:

- Because buses use the road as a drop-off zone, they must wait on the road until all passengers have alighted, creating long queues and traffic congestion.
- Buses enter the terminal empty. Buses of small companies use the terminal as a parking space, whereas large companies such as Hanif, Shyamoli and Saudia have their own

workshops. Repair and washing of small companies' buses therefore take place on the terminal premises, leaving them untidy and chaotic.

- There are no separate facilities for bus crews, terminal staff and passengers.
- Because no facilities are provided for drivers and conductors, they must rest inside their buses after long journeys, which harms their physical and mental health.
- Arriving passengers have no reason to enter the terminal or use the waiting room; instead, they wait on the road for feeder services such as rickshaws, CNG auto-rickshaws, local buses and taxis, which compromises their safety and convenience.
- Because most departing passengers buy tickets outside the terminal, they board buses on the road. Most passengers also do not know where terminal facilities are located, and because these are far from the bus pick-up points, they do not use them.
- Although the waiting room has adequate seating, it is occupied by vendors and terminal staff, which discourages passengers from using it.
- Some passenger sheds are unusable because their surfaces are broken, and these spaces are now used for bus repairs; ticket counters have consequently been set up informally under the flyover near the terminal exit.
- Poor hygiene and cleanliness are the main problems of the toilets maintained by the City Corporation. Because there are no separate toilets for bus crews, who stay in the terminal for long periods and need them frequently, the Tk 5 charge per use is unaffordable for them; as a result, they urinate on the terminal premises.
- There is no information center, locker room or medical center in the terminal.

6.3 Functional arrangement of Gabtoli Bus Terminal

The flows of buses and passengers are integral to a terminal's functional system, and the use of facilities provided for passengers, bus crews and terminal staff depends on these flows. The flows at Gabtoli terminal, based on observation and FGD 02, are described below.

6.3.1 Flow of arrival buses and passengers:

- Because the offloading area lies outside the terminal, passengers alight on the highway on the opposite side from the terminal.
- The local bus station and feeder services are on the same side of the highway, so passengers can easily reach them and continue their journeys.
- After setting down passengers, buses enter the terminal by making a U-turn at Technical Mor; all buses use the same entry point (Appendix E).

- Buses of small companies park in the terminal until their next trip, whereas those of large companies go to their own workshops.
- After parking, crews go home if they have no further trip that day. Any washing or repair is done before they leave, sometimes by laborers hired on daily wages. If crews have a further trip, they rest in their buses.

FGD-02: Opinion Regarding Functional Arrangement of Gabtoli Bus Terminal Given by Passengers

FGD conducted in the waiting room of Gabtoli Bus Terminal reveals that they use feeder services like CNG, Bus to arrive in the terminal. As the ticket counters are at the passenger shed, they wait in the passenger shed and don't wait in the waiting room normally. That's why, most of the time, they don't use toilets provided in the terminal building. Some passengers use the toilet facility and must pay 5tk per use but it is not reasonable to them. For the short distance buses, they don't need to buy ticket to get in bus in front of the terminal. They are normally unloaded from buses at the opposite side of the terminal and use the feeder services which are available near the terminal. Due to difficulty of crossing road to get in the terminal, they don't want to use the facilities provided for them. Because of absence of information center and direction sign in the terminal, they get lost in the terminal which causes waste of their valuable time.

6.3.2 Flow of departure buses and passengers:

- Passengers arriving by feeder services enter the terminal through a separate entry near the ticket counters.
- Passengers buy tickets and wait for buses in the passenger shed; ticket counter staff serve as the main source of information.
- Some passengers wait in the waiting room and use the public toilet beside it.

- Passengers can use the shops and canteens in the terminal.

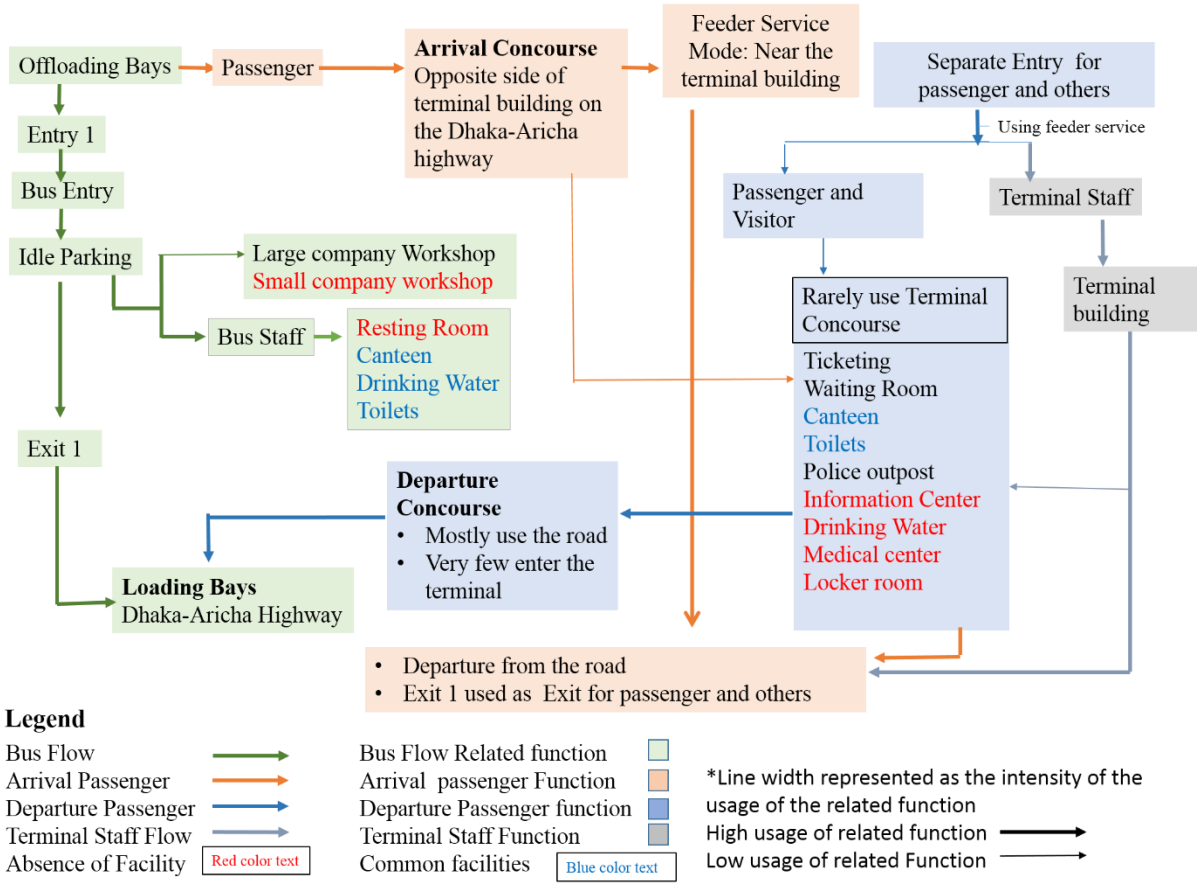


Figure 6.3: Functional Arrangement of Gabtoli Bus terminal

- Passengers board buses in the parking area. This process is informal: bus staff known as linemen guide passengers to their buses.
- Buses leave the terminal through the first exit. Buses parked on the Dhaka–Aricha highway pick up passengers outside the terminal; these passengers do not buy tickets but pay their fares directly to the crew on board.

6.3.3 Deficiencies of functional arrangement of Gabtoli bus terminal:

- Planning standards require loading and offloading to take place within the terminal, but at Gabtoli the offloading point is outside it. Passengers alight on the far side of the highway and leave by feeder services; since they would have to cross the road to reach the terminal building, most have no reason to use its amenities (Appendix E).
- Because there is no direct entry for buses, they must make a U-turn at Technical Mor, which frequently causes congestion there.

- There is no resting room for bus crews and terminal staff, who have to rest in buses and cannot rest properly, with consequences for their health.
- The ticket counters are close to the entry point, with passenger sheds in front of them, but the waiting room is far from the counters. Passengers therefore wait in the sheds, and the waiting room is underused.
- Because passengers are set down outside the terminal, they do not use its facilities. Few passengers and staff use the public toilets, even though the toilet in the waiting room is in good condition.
- Because there is no designated loading area, passengers have difficulty finding their buses and must rely on linemen.
- There is no information center. Passengers must obtain information from ticket counter staff, which sometimes leads to information gaps and long waits.

6.4 Functional arrangement of Mohakhali Bus Terminal:

Efficient flows of buses and passengers are necessary for a terminal to function smoothly. The functional arrangement of Mohakhali terminal and its deficiencies, based on observation and FGD 03, are described below.

6.4.1 Flow of arrival buses and passengers:

- Passengers alight on Tajuddin Ahmed Road and at the terminal entry point.
- Local buses and feeder services such as CNG auto-rickshaws and taxis are available in front of the terminal and at the exit, so passengers can easily continue their journeys.
- After setting down passengers, buses enter through the terminal's single bus entry and park in the parking area.
- All buses park within the terminal; Ena uses the loading bay as its parking space.
- Any washing or repair is done by crews before departure, sometimes with laborers hired on daily wages. Crews with a further trip that day rest in their buses.

FGD-03: Opinion Regarding Functional Arrangement of Mohakhali Bus Terminal Given by Passengers

FGD conducted in the waiting room of Mohakhali Bus Terminal reveals that they use feeder services like CNG, Bus to arrive in the terminal. Since most of the ticket counters are at the inside of the terminal building, they normally buy ticket from the ticket counters inside of the terminal and wait for the bus in the waiting room and use toilet facility frequently. They must pay 5tk per use of toilet which is more than reasonable for them. They are normally offloaded from buses at the outside of the terminal and use the feeder services which are available near the terminal. In some cases, they need to cross road to get in local buses with heavy bags and goods. In Mohakhali terminal they easily get access all the facilities and direction in the terminal.

6.4.2 Flow of departure buses and passengers:

- Passengers arriving by feeder services enter the terminal near the ticket counter area (via Entry 1 or Exit 1).
- Passengers buy tickets and wait for buses in the waiting room, where they can watch television. The information center provides information on arrival and departure times and the availability of buses.
- Passengers can use the shops and canteens in the terminal.
- Passengers board buses at the terminal exit, guided by bus staff.
- Buses leave through the exit point. At Mohakhali, all buses pick up passengers inside the terminal, although some passengers do not buy tickets and instead pay their fares directly to the crew (Appendix E).

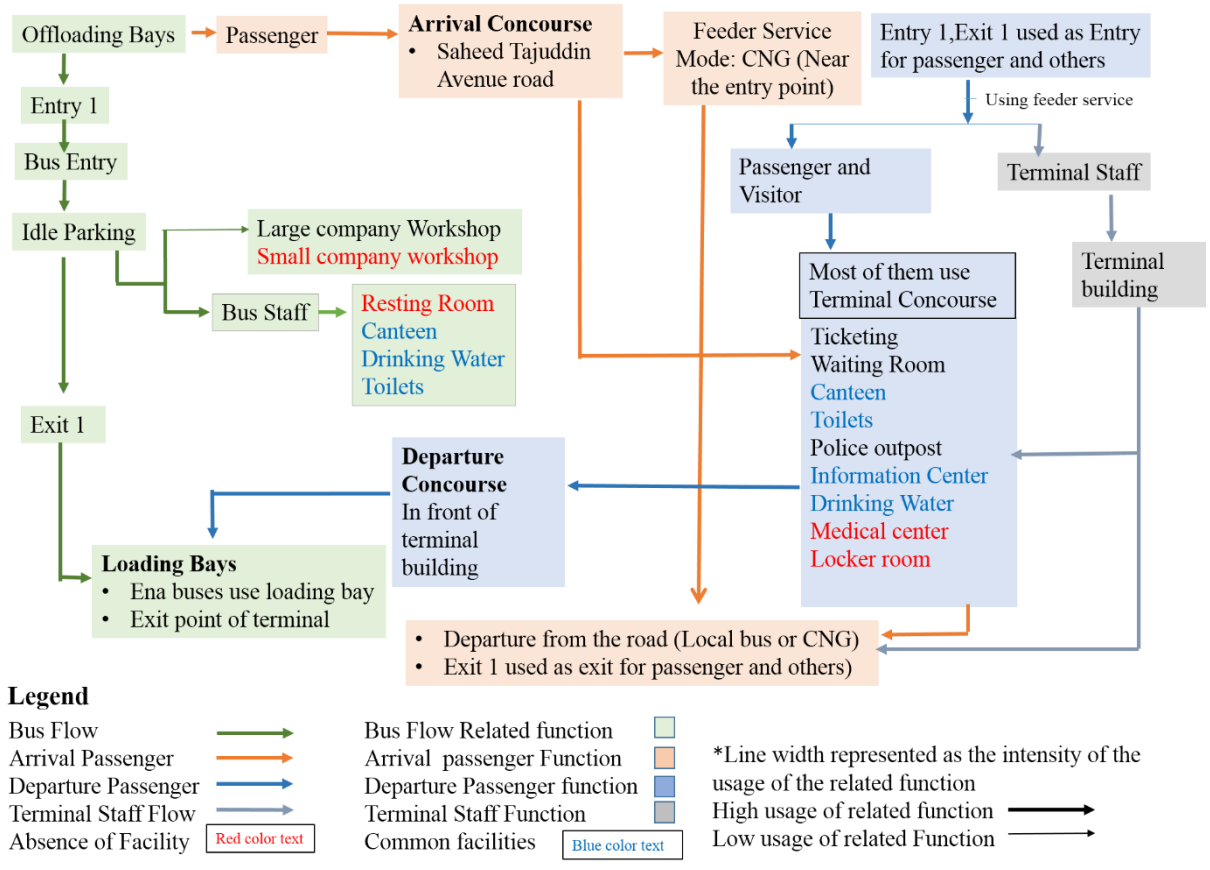


Figure 6.4: Functional Arrangement of Mohakhali bus terminal

6.4.3 Deficiencies of functional arrangement of Mohakhali bus terminal:

- Because Ena buses occupy the loading bay for parking, passengers have to board near the exit point.

- Departing buses must cross Tajuddin Ahmed Road, causing congestion, and long queues form near the terminal entry.
- There is no resting room for bus crews and terminal staff, who must rest in buses and cannot rest properly.
- There are no separate facilities for passengers, bus crews and terminal staff.
- The information center is not always staffed, so passengers cannot always obtain information there.
- Parking for feeder services is insufficient; CNG auto-rickshaws park at the terminal exit and obstruct bus movements (Appendix E).

6.5 Comparative summary

Across the three terminals, the deficiencies follow a common pattern when set against the standard arrangement in Figure 6.1. First, the terminal's core passenger function—boarding and alighting—has moved to the roadside: at Sayedabad and Gabtoli, most passengers board and alight on adjacent highways, and at Mohakhali the drop-off lane has been taken over for parking by a single operator. Second, the terminals function as depots: buses of small companies are parked, washed and repaired on the premises, crowding out the circulation space on which the standard arrangement depends. Third, facilities designed for passengers and crews are either absent (resting rooms, information desks, signage) or present but under-used, because they are not located on the paths that passengers actually follow. Mohakhali, where ticket sales are concentrated inside the terminal and an information desk operates, performs comparatively better on several of these criteria despite being the smallest of the three sites. The deficiencies therefore cannot be explained by site constraints alone. Chapters 7 and 8 examine the stakeholder relations and managerial practices that sustain them.

This chapter identifies the internal and external stakeholders of the three terminals, describes the management roles each performs, and analyzes the distribution of power and influence among them.

7.1 Internal and external stakeholders of bus terminal:

Internal stakeholders are directly involved in the day-to-day management and functioning of the terminals; external stakeholders are not directly involved but influence, or are influenced by, their management (Appendix E). The internal and external stakeholders identified are listed in Table 7.1.

Table 7.1: Internal and External Stakeholders of the Study

Internal Stakeholder	**External Stakeholder**
Dhaka North City Corporation (DNCC)	Local Politician
Dhaka South City Corporation (DSCC)	Regional Transport Committee (RTC)
Bus Owners Association of Three Terminals • Bangladesh bus truck owner association,Gabtoli Bus Terminal • Bangladesh Sarak Poribohon Malik Somiti, Mohakhali Bus Terminal • Sayedabad Bus malik Somiti	Dhaka Metropolitan Police (DMP)
Bus Terminal Labor Union of Three Terminals • Dhaka Zilla Janbahon Sromik Union, Gabtoli Bus Terminal • Dhaka Zilla Bus Minibus Sarak Poribohon Sromik Union (Reg No.2415), Mohakhali Bus Terminal • Dhaka Mohanogor Sorok Poribohon Sromik Union (Reg No.-2195), Sayedabad Bus terminal	Ministry of Local Government, Rural Development and Co-operatives (LGRD&C)

Collector of Each Terminal	Dhaka Transport Coordination Authority (DTCA)
Driver, Conductor, Ticket Counter Staffs	Bangladesh Road Transport Authority (BRTA)
Informal Maintenance Worker	Bangladesh Sarak Paribahan Sromik Federation

7.2 Management done by various stakeholders:

Internal and external stakeholders participate in management in different ways. Although each has prescribed procedures, these are frequently not followed; activities take place outside formal rules and regulations, and informal negotiation has become routine. The managerial activities of each stakeholder are described below.

7.2.1 City Corporation Official:

The Mayor and the Chief Executive Officer are the approval authorities of the two city corporations, DNCC and DSCC. The General Manager (Transport) heads the bus terminal department and is responsible for revenue collection, cleaning, and the provision of safe and secure services to passengers. The General Manager also decides on public works in the terminals, such as installing lighting and constructing toilets and

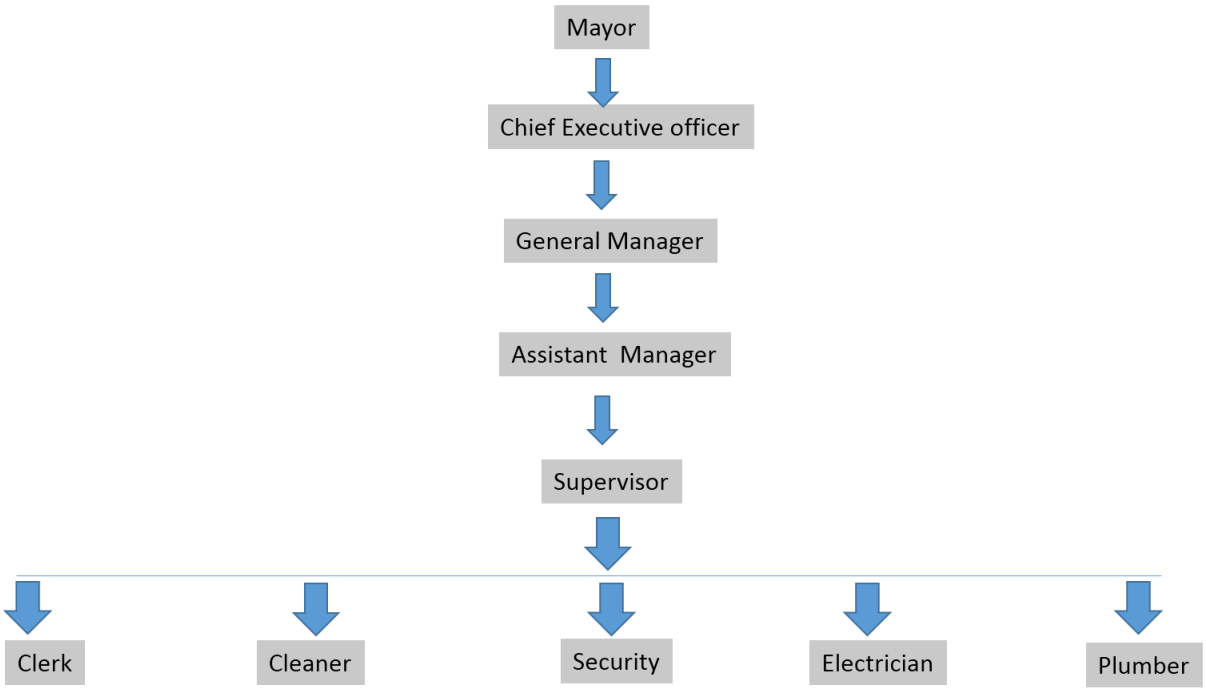


Figure 7.1: Hierarchy of Authority of Bus terminal (City Corporation)

internal roads. The Assistant Manager is directly involved in day-to-day terminal management and works mainly from the terminal office (Appendix K).

The Supervisor assists the Assistant Manager and performs all of his duties in his absence. Together, they collect daily tolls from buses, toilets and shops and keep daily records of bus entries and exits. They are also responsible for maintaining and repairing amenities such as the waiting room, toilets, lighting and internal roads, and for evicting hawkers who occupy the terminal area and internal roads. Clerks, electricians, security guards, plumbers and cleaners support the smooth running of the terminal (Figure 7.1).

7.2.2 Bus Terminal Owners Association:

Although each terminal has its own owners' association, their organizational structures are similar (Figure 7.3). Their main responsibilities are to:

- Represent bus owners' interests relating to the terminal;
- Grant permission to new buses to use the terminal; and
- Hold regular meetings with government agencies.



Figure 7.2: Bus terminal labor Union



Figure 7.3: Bus terminal owner association

Source: Field

Owners of buses using the terminal are the main members of the association, and a bus owner who is not a member cannot park buses or pick up passengers at the terminal. Owners of large companies such as Hanif and Shyamoli hold senior positions in the association, and political influence is a major factor in obtaining such positions. The Chairman takes the major decisions, in coordination with the Vice Chairman, General Secretary, Assistant Secretary, Treasurer and executive members (Appendices J and K). Although each office-holder has defined responsibilities, most do not fulfill them because they are occupied with other businesses; office

employees handle daily activities on their behalf. Association members come to the office to take major decisions only during political events or emergencies such as strikes or hartals.

7.2.3 Bus Terminal Labor Union:

Each terminal has its own labor union (Figure 7.2), whose main stated responsibility is to improve workers' livelihoods. Workers involved in terminal activities—drivers, conductors, helpers and mechanics of various kinds (engine, electrical, wheel, glass, paint and grease mechanics, locally called mistri)—are all members of the union and can bring complaints to it (Appendix I). The office-holders—President, Vice President, General Secretary, Joint Secretary, Sports Secretary, Treasurer, Office Secretary, Publication Secretary and executive members—rarely come to the union office despite their responsibilities (Appendices J and K); union employees receive complaints, pass them on to members and carry out other responsibilities on their behalf. Union representatives attend meetings with the owners' association or the government during emergencies or when new projects are implemented. The terminal labor unions are affiliated to the Bangladesh Sarak Paribahan Sromik Federation, which resolves internal conflicts.

7.2.4 Collector, Driver, Conductor and informal worker:

Because of staff shortages, the City Corporation engages collectors drawn from the labor union to collect tolls on its behalf; collectors often collect tolls from drivers or conductors. Informal workers are also hired to fill shortages of cleaners, security guards, plumbers and other staff.

7.2.5 Other Government Organization:

Other government organizations—DMP, BRTA, RTC, DTCA and LGRD&C—are indirectly involved in terminal management, as described below.

- **DMP:**

Traffic police control traffic around the terminals and help buses enter and leave (Figure 7.5). When buses stop at the roadside to pick up passengers, the police signal them to move on and fine those that do not comply (Appendix K). Illegal activities such as drug dealing sometimes occur inside the terminals, with passengers carrying drugs to exchange with dealers; the terminal police outposts deal with such cases strictly (Figure 7.4).

- **BRTA:**

BRTA is responsible for managing and monitoring buses at the terminal, including checking their licenses and fitness.



Figure 7.4: Police outpost in the bus terminal



Figure 7.5: Traffic police working outside the terminal

Source: Field

- **RTC:**

The Regional Transport Committee (RTC), which issues route permits in coordination with BRTA, grants permission to new buses to operate from the terminal. Representatives of the owners' association sit on the committee.

- **LGRD&C:**

The city corporations fall under the Local Government Division of the Ministry of Local Government, Rural Development and Co-operatives (LGRD&C), which appoints all terminal employees.

- **DTCA:**

The Dhaka Transport Coordination Authority (DTCA) is responsible for coordinating Dhaka's transport system. Because terminals are focal points of that system, DTCA incorporates them in its plans.

7.3 Exercise of power and relationships among stakeholders

The power–influence matrix and the Venn diagram are used to examine how stakeholders exercise power and how they relate to one another.

7.3.1 Power-Influence Matrix of stakeholders of bus terminals:

The power–influence matrix is a technique of stakeholder analysis that shows how stakeholders vary in power and influence. Here, power refers to a stakeholder's level of formal authority, and influence to how actively it shapes decisions in practice. Figure 7.6 shows the power and influence of the stakeholders involved in managing Dhaka's bus terminals.

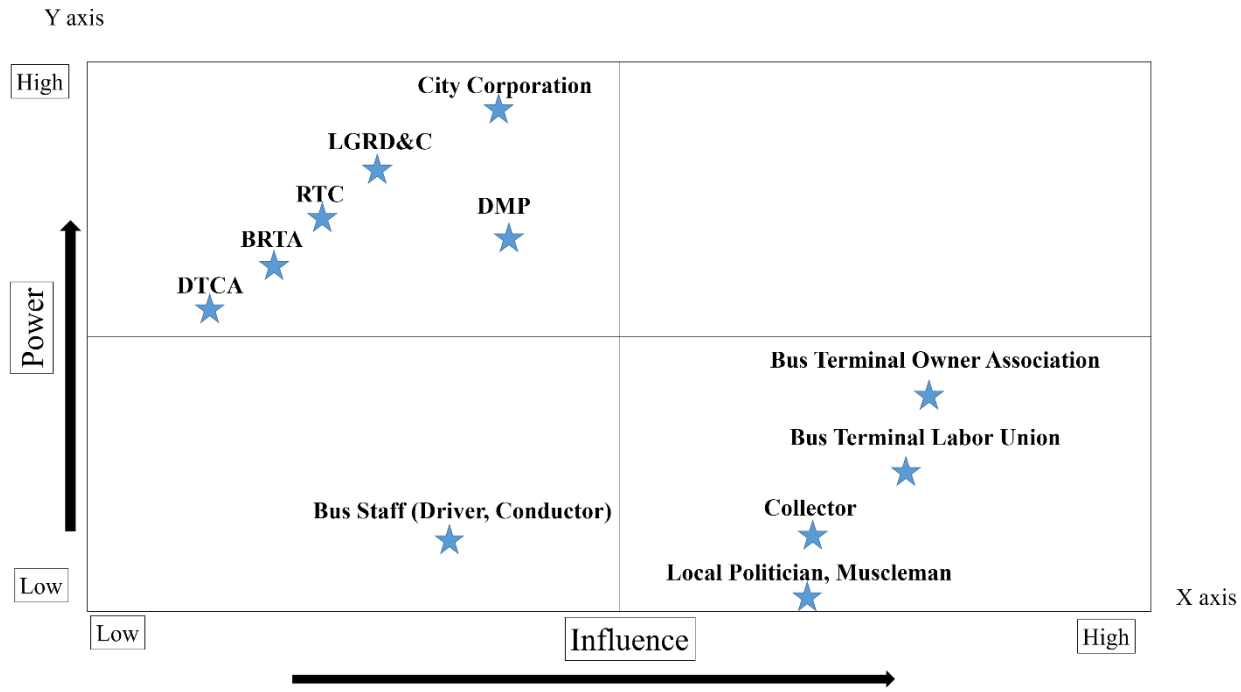


Figure 7.6: Power influence matrix

Figure 7.6 shows that no stakeholder combines high formal power with high influence over terminal management. One would expect those with the highest authority to exercise the most influence. However, government bodies such as the city corporations and LGRD&C, despite their high formal authority, cannot effectively influence terminal management. Conversely, the bus owners' associations, which rank low in the formal hierarchy, exert the greatest influence and shape the functional and managerial system of the terminals in their own interest. This inversion—formal authority without effective control, and effective control without formal authority—is the central finding of the stakeholder analysis and is examined further in Chapters 8 and 9. In Helmke and Levitsky's (2004) terms, the owners' associations and unions operate informal institutions that substitute for the formal rules of terminal management: the formal hierarchy remains in place on paper, but the rules that govern access, parking and revenue are effectively set elsewhere.

7.3.2 Venn diagram of stakeholders of bus terminals:

The Venn diagram (Figure 7.7) shows the relationships among stakeholders, distinguishing direct (internal) from indirect (external) stakeholders. It analyzes the interactions between the stakeholders and the city corporations, the formal authorities of the three terminals, along three dimensions: influence, and the direction and intensity of interaction. The main observations are as follows:

- Of all stakeholders, the bus owners' associations exert the greatest influence over the management of all three terminals. All owners of buses using a terminal are members of its association, which sets bus fares and schedules and, according to informants, illegally appropriates a large share of the revenue from tolls and other fees (Appendix E).
- Although DNCC and DSCC are officially responsible for managing the terminals, in practice they cannot control their management. For example, the City Corporation has fixed the bus parking fee at Tk 40, but the amount actually charged is higher. The City Corporation therefore ranks only third in terms of control over the terminals.
- Interaction between the labor unions and the City Corporation intensifies before major occasions such as Eid-ul-Fitr. Because collectors drawn from the unions collect tolls and other fees, the unions retain a portion of the revenue before it reaches the authority. Local politicians and 'musclemen' are involved with both the labor unions and the owners' associations; in effect, local politicians, owners' associations and labor unions together control the terminals. The unions operate mainly on the ground, and relations between them and the owners' associations are currently good.
- The Supervisor and Assistant Manager, whose offices are in the terminal building, manage the terminals on behalf of the City Corporation. The Supervisor communicates daily with collectors over tolls and fees from hotels and shops and appoints informal staff on daily wages; interaction between formal staff and the Supervisor is frequent.
- Bus crews, linemen and ticket counter staff are directly involved in, and in effect operate, the functional processes of the terminal. They interact with passengers hourly, but the communication is one-directional.
- DMP is responsible for passenger security in the terminals and checks vehicle documents near the terminals, and it therefore interacts frequently with the labor unions and owners' associations. Passengers regularly bring complaints to nearby DMP offices.

- BRTA and RTC deal with route permits and bus documents; BRTA normally communicates with the City Corporation's General Manager regarding the terminals. Because two members of the RTC represent the bus owners' association, the association interacts frequently with the RTC to obtain route permits (Appendix E).

- The Bangladesh Sarak Paribahan Sromik Federation rarely interacts with the terminal labor unions.

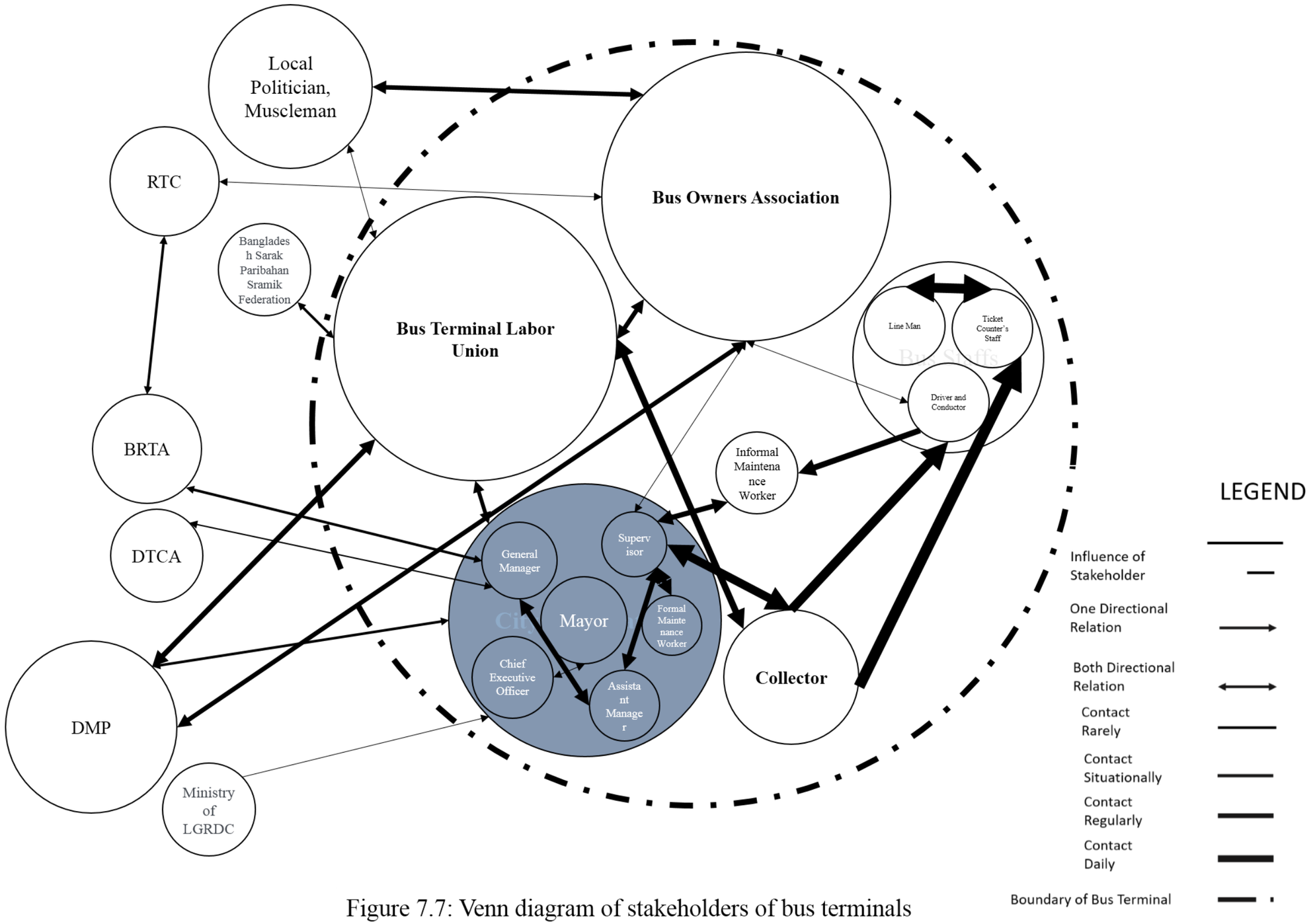


Figure 7.7: Venn diagram of stakeholders of bus terminals

The ineffective management of the terminals gives rise to a range of problems. This chapter traces the evolution of the management system and analyzes the main managerial problems, their causes and their effects.

8.1 Evolution of the management system of bus terminal:

There is a long history behind the present condition of the three terminals. Their construction and management have changed gradually over time, and management has become increasingly informal (Appendix E). Tracing this history helps to explain the current managerial problems. Table 8.1 presents a timeline of the three terminals.

Table 8.1: Timeline of the Inter District Bus Terminals of Dhaka

Time	Construction Work	Administrative Change
1945	Establishment of Gabtoli Bus Terminal	
1950's	Establishment of Sayedabad Bus Terminal	
1984	Establishment of Mohakhali Bus Terminal	
1985		Appointment of Staffs for the terminals by LGRD&C
1990's		Authorizing power handover to Dhaka City Corporation
		Political involvement in management control
October, 2003	Development work under DUTP project started	Management control of three bus terminals taken by Rora and Diba private organization
March, 2004		Cancellation of the contract with Rora Enterprise at Sayedabad and Mohakhali Bus Terminal
2004-		A court case filed by DCC against Rora Enterprise
		Case still going on
June, 2004		Panama Enterprise took management control of Sayedabad and Mohakhali Bus Terminal
2005	Development work finished	Appointment of city corporation officials for the bus terminals
2007		Management control power taken by Dhaka City Corporation again from Panama and Diba enterprise

		Political involvement abated in the terminals due to Caretaker Government
2009		Political power involved in the terminals again after regular Government obtained power

Gabtoli terminal was established first, in 1945, to serve the northern districts, followed by Sayedabad terminal for the southern and eastern districts before 1960, and Mohakhali terminal in 1984. The following year, the local government appointed staff for each terminal, and in 1990 Dhaka City Corporation (DCC) assumed authority over all three.

In 2003, the authority invited tenders to hand over management of the terminals to private organizations. Rora Enterprise obtained the management of Sayedabad and Mohakhali terminals, and Diba Enterprise that of Gabtoli. After Rora Enterprise breached its agreement, DCC cancelled the contract and filed a case against it. Panama Enterprise then managed the two terminals but also failed to recover the expected revenue. Meanwhile, under the Dhaka Urban Transport Project (DUTP), the terminals were upgraded structurally between 2003 and 2005: terminal buildings, passenger sheds and ticket counters were constructed. In 2005, the local government appointed new City Corporation officials to the terminals.

In 2007, the City Corporation resumed management of the terminals. Political involvement in terminal management abated under the caretaker government of 2007–2008 but returned in 2009, when an elected government took office, and has since extended to revenue collection. The case against Rora Enterprise remains unresolved.

8.2 Problems related to managerial issue of bus terminals:

Because terminal management is heavily influenced by politics, the system cannot run smoothly. The resulting problems fall into two groups: administrative problems and facility-related problems.

8.2.1 Administrative issues

Administrative problems arise when administrative work is carried out informally. They are described below.

- **Mismanagement of the toll collection system**

Tolls collected from bus parking, shops, toilets, porters (kuli mojuri) and ticket counters are the main source of terminal revenue, and the amount varies with the number of buses using each terminal. Buses are required to pay Tk 40 each time they enter a terminal but in practice pay only once a day, even if they enter two or three times. According to informants, the City Corporation receives Tk 117,000 per day from Gabtoli, an average of Tk 65,000 per day from Sayedabad and Tk 50,000 per day from Mohakhali (Appendix E). This direct collection by the authority is known as 'khas collection'. The causes and effects of the problem are shown in Figure 8.1.

Causes:

Because tenders cannot be invited while the case against Rora Enterprise remains unresolved, toll collection has reverted to the City Corporation. Lacking sufficient staff, the City Corporation recruits collectors from the labor union, and a porters' leader (kuli mojuri sardar) also collects money from porters at the terminal. If tenders could be invited, a private contractor would take over these roles. In the absence of a contractor, the khas collection process lacks accountability and transparency because of the involvement of the collectors and the porters' leader. Informants identified political influence as the main cause of informal toll collection (Appendix E): because the whole transport sector is subject to such influence, control of bus terminals gives political leaders opportunities for rent-seeking and extortion.

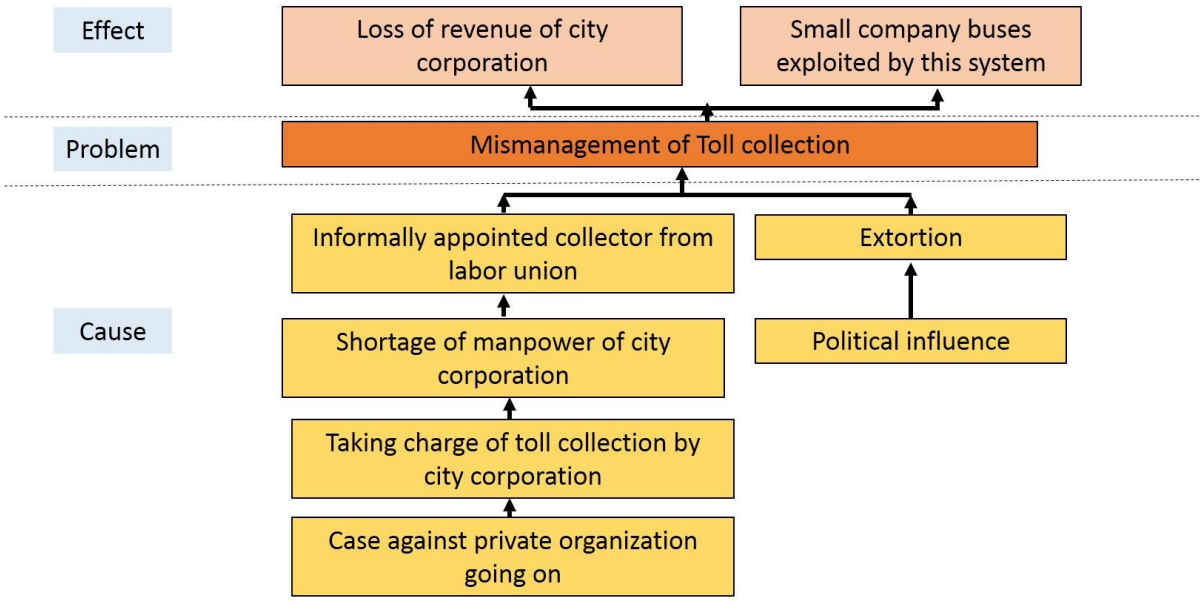


Figure 8.1: Cause effect diagram for mismanagement of toll collection

Effects

Collectors working under the labor union charge buses Tk 400–500 per day for parking. Buses of small companies, with lower revenues, find this extra payment particularly burdensome. Meanwhile, the collectors remit only a portion of the fixed daily toll to the City Corporation, which consequently loses revenue every day.

- **Shortage of staff**

Neither city corporation has enough staff to manage the three terminals. Each terminal requires around 30–40 staff, including cleaners, plumbers, security guards, electricians and clerks, but Sayedabad, Gabtoli and Mohakhali have only 12, 18 and 8 respectively (Appendix E). The causes, sub-causes and effects of the shortage are shown in Figure 8.2.

Causes:

In 1985, LGRD&C appointed City Corporation officials and terminal staff to manage the terminals, and in 2005 it appointed new officials to all three terminals. However, no new terminal staff have been appointed since 1985; over this long period, staff have retired or died, leaving all three terminals understaffed (Appendix E).

Local political influence is a parallel cause. Local politicians seek to influence terminal management and, according to informants, oppose new appointments that might obstruct their illegal activities.

Effects:

Because of the staff shortage, the City Corporation cannot maintain facilities regularly. Passenger sheds and ticket counters are deteriorating, parking and repair areas are not cleaned, and the terminal environment is poor (Appendix J). The lack of security staff has gradually worsened security conditions.

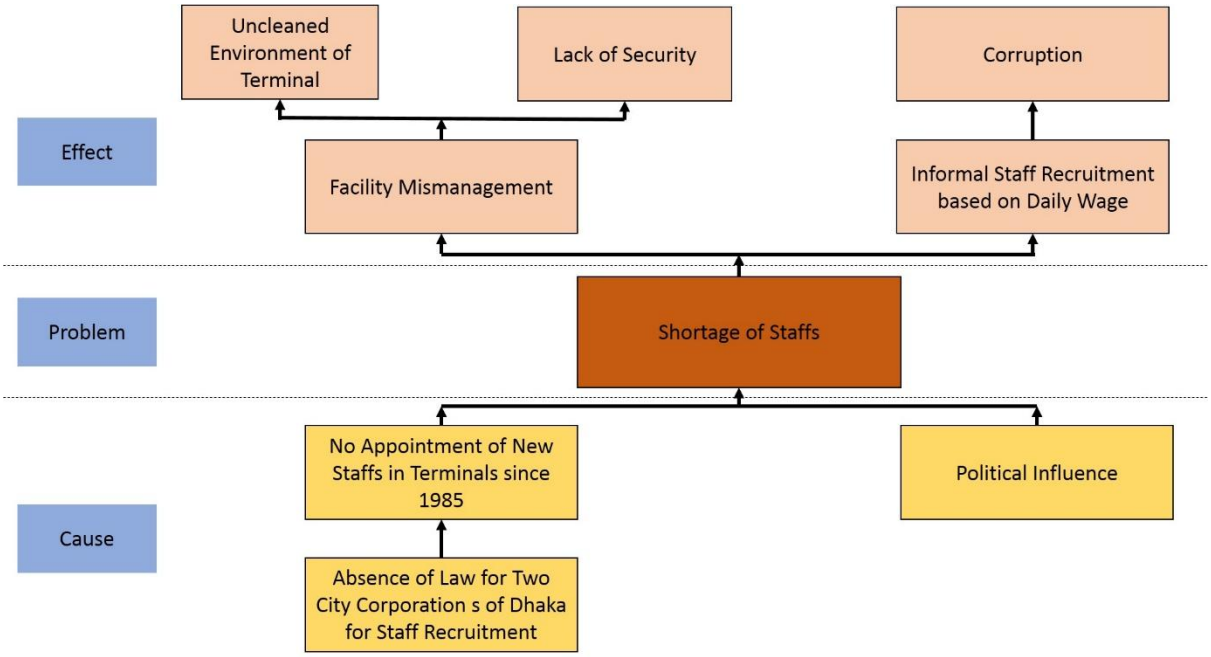


Figure 8.2: Cause effect diagram for shortage of staff

To meet its staffing needs, the City Corporation appoints local workers on daily wages (Appendix E). Because there is no formal recruitment process, this practice creates opportunities for corruption in toll collection and maintenance, and maintenance remains irregular despite the daily appointment of informal staff.

8.2.2 Facility-related issues

Several amenities provided in the terminals are not used effectively because of managerial problems. These are described below.

- **Mismanagement of bus parking**

Parking demand exceeds supply in all three terminals, and owners must negotiate with the owners' association to obtain parking space. The Dhaka–Sylhet highway in front of the Sayedabad exit is frequently congested, although only briefly, because traffic police manage the flow. Shaheed Tajuddin Ahmed Sarani in front of Mohakhali terminal becomes congested with arriving buses, which take a long time to reach their parking spaces. At Gabtoli, because buses turn around at Technical Mor, congestion occurs there rather than in front of the terminal (Figure 8.3).

Causes:

To operate a bus, an owner must obtain a route permit from the RTC. To do so, the owner must first become a member of the owners' association and pay it a fee. Because the RTC includes representatives of the association, the association acts as the intermediary between permit applicants and the committee. As a result, owners can obtain route permits without having a workshop or garage in which to park their buses (Appendix E). Moreover, owners treat the terminal as a depot because most other owners do the same. This generates excess parking demand and the mismanagement of parking in the terminals. The mechanism is a form of regulatory capture (Stigler, 1971): the committee that regulates entry into the bus market includes representatives of the incumbents it regulates, and the association acts as a gatekeeper that converts its control over permits into membership fees and influence. Similar gatekeeping by operator associations and unions has been documented in other cities of the Global South (Cervero & Golub, 2007; Agbiboa, 2018).

Effects:

During peak hours, many buses enter and leave the terminal at the same time. Because of excess demand, internal roads are blocked by parked buses, and the mismanagement of parking also causes congestion outside the terminal (Appendix J).

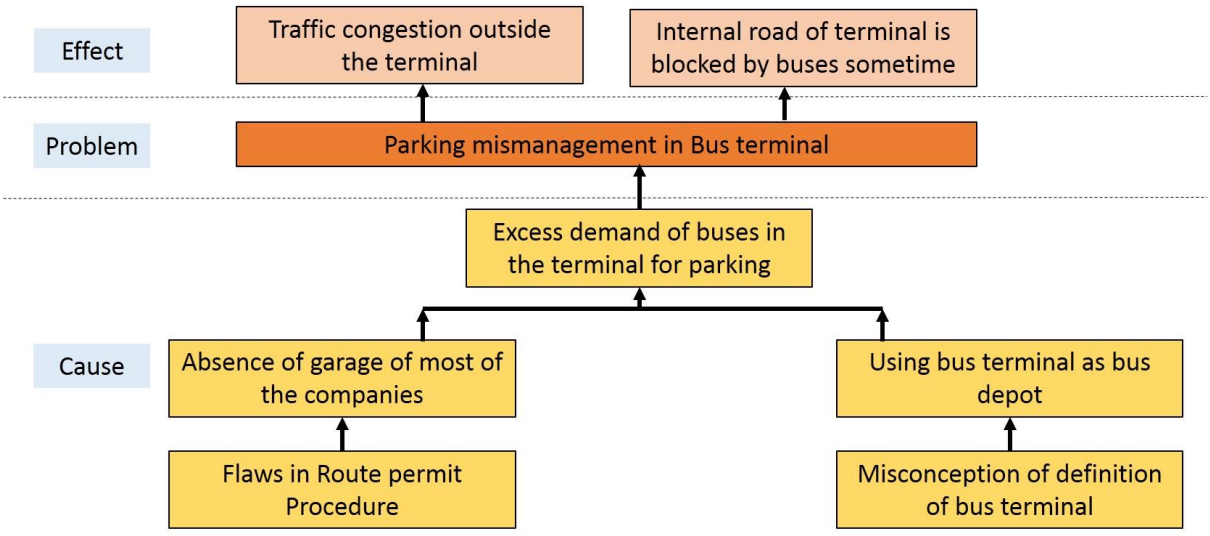


Figure 8.3: Cause effect diagram for parking mismanagement

- **Ticket counters outside the terminal**

Ticket counters for inter-district buses have spread across Dhaka, allowing passengers to board buses without coming to the terminal (Figure 8.4).

Causes:

Many bus companies operate on the same routes, and competition among them is intense. Companies hold counter space inside the terminals but do not use it, because passengers do not come there and instead buy tickets from external counters; to attract more passengers than their rivals, companies set up counters outside the terminal area. Owners' association leaders and owners also set up counters informally across the city: around Technical Mor, Darus Salam, Shyamoli, Kalyanpur, Asad Gate, Mirpur Mazar Road and Mirpur Sections 1 and 2 for Gabtoli; under the Mayor Mohammad Hanif Flyover and at Fakirapool, Motijheel, Plassey Square, Maniknagar, Basabo, Malibagh, Arambagh, Kamalapur, Khilgaon, Rampura, Badda, Jatrabari, Postagola, Jurain, Signboard and on Chattogram Road for Sayedabad (Karmakar & Alam, 2016); and at the Airport, Uttara and Abdullahpur for Mohakhali. Traffic police also allow buses to stop at roadside counters in exchange for bribes.

Effects:

Buses stopping indiscriminately at the roadside obstruct pedestrians and other vehicles and cause congestion. The City Corporation also loses revenue, since it can collect tolls only from counters inside the terminal (Appendix J).

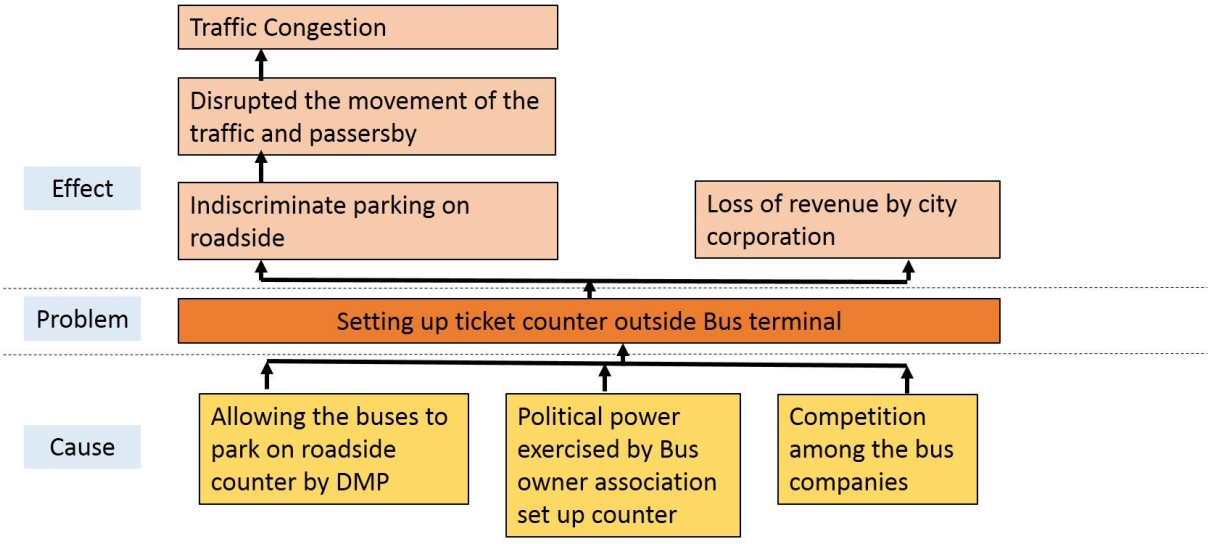


Figure 8.4: Cause effect diagram for outside ticket counter

- **Absence of resting rooms in the terminals**

Although resting rooms are important, none of the three terminals has one for bus crews or terminal staff. The reasons and effects are shown in Figure 8.5.

Causes:

Resting rooms are a prime concern of the terminal labor unions but were not included in the design of the terminals, and they were overlooked again in the structural improvement project.

The City Corporation's position is that, because a terminal may not be used as a depot and every bus company must have its own workshop and garage, crews should rest at the workshop or garage; since buses are not supposed to stay long in the terminal, a resting room is not considered necessary. This disagreement is a source of conflict between the labor unions and the City Corporation (Appendix E).

Effects:

Without resting rooms, drivers and conductors must rest in their buses and cannot rest properly. Their health is deteriorating as a result, and fatigue may contribute to road accidents (Appendix J).

The Motor Vehicles Ordinance, 1983 requires drivers to rest after every five hours of driving (Appendix E). Because the terminals lack resting rooms, drivers and helpers cannot rest properly after their trips, and the Ordinance is not effectively implemented.

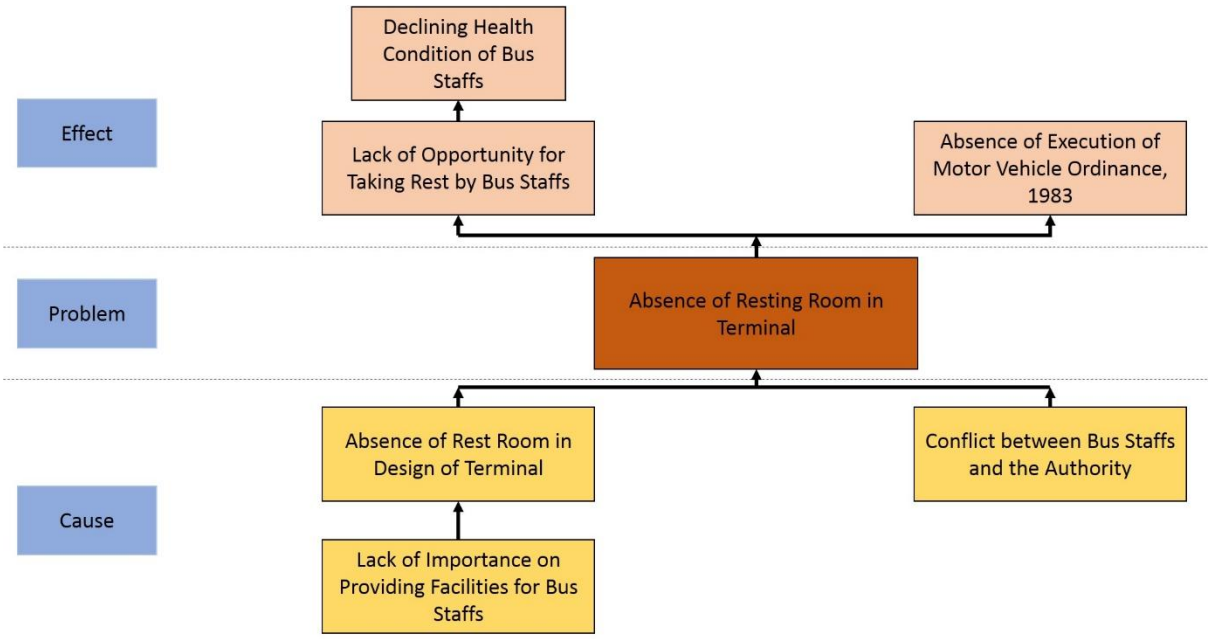


Figure 8.5: Cause effect diagram for absence of resting room

8.2.3 Other managerial issues

Because WaterAid constructed some of the toilets, it has the authority to collect user fees for them. Fees for the other toilets are nominally collected by the City Corporation, but the collector from the labor union retains the entire amount. Users pay Tk 5 per visit. Although this is a small sum, frequent users such as workers in the terminal find it difficult to pay each time; as a result, they urinate on footpaths and in unused areas, making them dirty and unhygienic (Figure 8.6). Similarly, bus washing costs around Tk 200–500, and the labor union receives all of this money.



Figure 8.6: Footpath used by bus staffs instead of using toilet

Source: Field

Chapter 9: Major Findings, Discussion and Recommendations

This chapter summarizes the major findings from the analysis of the terminals' functional arrangement and management, discusses them in relation to the literature, and presents recommendations.

Functional arrangement and facilities

1. At Mohakhali terminal, ticket counters are located only around the terminal building. Because there are no counters outside, passengers must come to the terminal and therefore use its facilities. By contrast, many counters operate outside Sayedabad and Gabtoli terminals, so passengers there do not need to come to the terminal.
2. Mohakhali terminal has a passenger loading area that passengers use properly, whereas the other terminals have no designated loading area, so passengers wait and board on the road.
3. Mohakhali terminal has an information booth that helps passengers find facilities; the other two terminals have none, so passengers must ask strangers for directions.
4. Feeder services are readily available at Mohakhali terminal, allowing passengers to transfer easily after alighting; at Sayedabad and Gabtoli, feeder facilities are not used properly.
5. The entry and exit ramps of the Hanif flyover inside Sayedabad terminal help buses avoid congestion near the terminal.
6. Some facilities are common to all terminals: police outposts, petrol pumps, repair shops, WaterAid toilets, adequate seating and sufficient canteens.
7. Drainage at Sayedabad terminal is poor, and waterlogging is a common problem that degrades the terminal environment.
8. All three terminals front national highways, which facilitates the movement of inter-district buses.
9. Problems common to all terminals include the poor condition of toilets managed by the City Corporation, insufficient parking space, the lack of separate entry and exit facilities for passengers, the lack of signage and the absence of resting rooms.

Management and governance

1. Each terminal has a bus owners' association and a labor union, representing owners and workers respectively, and the two maintain good relations.

2. Regular elections are held in the owners' associations and labor unions, which helps to limit internal conflict.
3. Management is ineffective in all terminals because the labor unions and owners' associations have greater influence over decision-making than the City Corporation. The unresolved case against Rora Enterprise has prevented the involvement of a private operator.
4. Staff numbers are insufficient for managing the terminals, so the City Corporation recruits staff informally on a daily basis. It has no legal authority to appoint new staff, and LGRD&C, which does, has been reluctant to do so.
5. Extortion from owners and workers is common in all terminals because of political influence.

SWOT analysis of terminal facilities and management

SWOT analysis identifies the internal attributes (strengths and weaknesses) and external attributes (opportunities and threats) of an organization or institution (Investopedia, 2019). Figure 9.1 summarizes the facilities and management of the terminals in SWOT form.

Discussion: functional failure as a spatial outcome of informal governance

Read together, the findings suggest that the functional deficiencies documented in Chapters 5 and 6 are not primarily failures of design. Each of the most consequential deficiencies can be traced to a specific governance mechanism identified in Chapters 7 and 8. The use of terminals as depots follows from a route-permit process in which owners' associations act as gatekeepers and permits are issued without proof of depot space. Roadside loading and unloading follow from the proliferation of external ticket counters, which reflects intense competition among operators and is tolerated by enforcement agencies in exchange for informal payments. The neglect of passenger and crew amenities follows from a revenue system in which tolls are captured by collectors drawn from the labor unions, leaving the City Corporation without the staff or income needed for maintenance. In each case, the space of the terminal is organized around the interests of the actors who control it rather than around the requirements of passengers or the standards of planning guidance. The pattern closely resembles that described by Agbiboa (2018) in the motor parks of Lagos, where union-affiliated touts tied to political patrons collect levies and resist removal; the Dhaka case adds that such control is also visible in the physical layout and use of the terminal.

The comparison among terminals supports this interpretation. Mohakhali, the smallest terminal and the one with the most severe site constraints—a single entry and no departure lane—nonetheless performs comparatively well on several functional criteria, because, according to the observation data, ticket sales are concentrated inside the terminal and an information desk operates there. The deficiencies observed elsewhere are therefore not inevitable consequences of land shortage or congestion but depend on how operations are governed.

These findings are consistent with the view of urban informality as a mode of governance rather than an absence of regulation (Roy, 2005; Hackenbroch & Hossain, 2012). The formal framework—city corporation ownership, BRTA licensing and RTC route permits—remains in place, but its operation is shaped by informal institutions that, in Helmke and Levitsky's (2004) terms, compete with or substitute for the formal rules. The power–influence matrix captures this as an inversion of formal authority and actual influence. The timeline adds a temporal dimension: political involvement in the terminals declined under the caretaker government of 2007–2008 and returned in 2009, indicating that terminal governance is tied to the wider political settlement (Khan, 2010; Hassan, 2013) rather than to the capacity of any single agency. Reports since the change of government in August 2024 that control of terminal offices and associations passed from one party's affiliates to another's, while collections in the name of owners' and workers' associations continued (Prothom Alo, 2024; The Business Standard, 2024), point in the same direction. The case therefore extends the informality lens from settlements and services to transport infrastructure (Banks et al., 2020): the terminal is not an informal space but a formal facility that is governed informally. It also reverses the familiar image of informality as the domain of the poor; as in Hackenbroch and Hossain's (2012) account, and in contrast to Bayat's (2000) 'quiet encroachment of the ordinary', the encroachment here is organized and led by the powerful.

This interpretation has a direct policy implication. Physical upgrading alone has not resolved the terminals' problems: the DUTP works of 2003–2005 built new terminal buildings, sheds and counters, yet the facilities remain under-used because the arrangements that divert passengers and revenue elsewhere were left untouched. The Government's 2026 plan to relocate the inter-district terminals to the periphery and to separate depots from passenger terminals (The Business Standard, 2026) addresses a real problem identified in this study, but relocation risks reproducing the same pattern at new sites unless route permitting, revenue collection and terminal management are reformed at the same time. The recommendations below therefore address governance first and physical provision second.

Recommendations

International experience suggests that reforms of informally governed public transport succeed more often when they engage existing operators, associations and workers rather than seeking simply to displace them, and when they proceed incrementally (Kumar et al., 2021). The recommendations below therefore combine stronger public authority over the terminals with a defined, accountable role for organized operators and workers.

- The case against Rora Enterprise should be resolved quickly so that tenders can again be invited. A private operator working under a performance-based contract with the City Corporation, and insulated from political interference, could manage the terminals more effectively, since its contractual obligations would make it accountable and limit opportunities for non-transparent collection.
- Bus operators that bypass the terminals and collect passengers through roadside counters should be required to use the terminals and to board passengers only at counters inside them. Traffic police should strictly enforce the rules against roadside counters and fine operators who

Strength	Weakness
• Ticket counters inside the terminal building in Mohakhli bus terminal • Usage of specific passenger loading area properly in Mohakhli bus terminal • Information booth available in Mohakhli bus terminal • Availability of feeder service in Mohakhali bus terminal • Enough sitting arrangements for passengers in all terminals • Enough canteen available for passengers and staffs of terminals • Proper maintenance of Wateraid toilet in all terminals • Availability of bus owner association in all terminals • Availability of bus terminal labor union in all terminals • Good relationship between bus owners association and labor union • Presence of election system in bus owner association and labor union	• Improper drainage system in the Sayedabad Bus Terminal. • Bus bays, occupied by Ena Bus Companies in Mohakhali Bus Terminal illegally • Undesignated arrival, departure lane and passenger loading and unloading area in Gabtoli and Sayedabad Bus Terminal • Absence of present document of master plans for the terminals to the authority. • Higher power of Labor Union and Bus Owners Associations than city corporation in decision making process of the terminal. • Insufficiency of staffs for managerial work of the terminals • Daily recruitment of new staffs by city corporation informally. • Lot of informal activities in the terminals • Poor condition of public toilet managed by city corporation authority in all three terminals. • Insufficiency of parking space in all three terminals. • Lack of entry and exit facilities in the terminals • Poor quality of food of the hotels in the terminals • Lack of signage to direct passengers in all three terminals • Source of water far away from bus washing place. • Absence of resting room in the terminals
Opportunity	**Threat**
• Availability of exit and entry ramp of Hanif flyover in Sayedabad bus terminal • Police station available in all terminals • Controlled the movement of terminal buses by traffic police around all the terminals • Repair shop available in all terminals • Petrol pump available outside the terminals • National highway in front of all the terminals	• Lots of ticket counters outside of Gabtoli and Sayedabad Bus Terminal • Unfinished case against Rora Enterprise • Absence of law of LGRD&C to recruit new staffs by city corporation in the terminals • Traffic congestion outside the terminals • Political influence in the terminals for extortion.

Figure 9.1: SWOT analysis of management and facility related the bus terminals

load and unload passengers at unauthorized locations. The City Corporation should also prohibit owners from setting up ticket counters outside the terminals.

- The required terminal staff should be appointed. Appointments should be regularized by LGRD&C, which alone has the authority to make them, on the basis of need; alternatively, delegating this authority to the City Corporation would allow staffing to respond more effectively to local needs.
- BRTA should apply strict criteria when allocating route permits, requiring owners to show proof of off-terminal depot space for their buses. Owners' association representatives may sit on the RTC, but they should not be able to use political influence to obtain permits, which should be issued on the basis of demand.
- A resting room for drivers should be provided in each terminal, since rest after long journeys is essential. The conflict between the City Corporation and the labor unions over resting rooms should be resolved; even if the City Corporation is reluctant, space should be provided, because small bus companies cannot afford such space near the terminals.
- Many functional problems would be resolved if these managerial problems were addressed. Without political protection, ticket counters would not be set up outside the terminals; passengers would buy tickets at the terminals; buses would load and unload passengers at designated places; and passengers would use amenities such as the waiting rooms, toilets and passenger sheds.
- Consolidating the operators that use each terminal into a small number of companies (fewer than five, as suggested by an informant) would reduce the number of ticket counters and remove the need for counters outside the terminal; passengers would then come to the terminal to buy tickets and use its facilities (Appendix E).
- With fewer companies, the cost to each company of idle parking in the terminal would increase, reducing the tendency to park idle buses there (Appendix E).
- Terminal management as a whole would improve if the associations, the city corporations and government agencies worked in a coordinated way (Appendix E). One option is a joint terminal management committee chaired by the City Corporation.
- A passenger association should be established to represent passengers' interests in meetings on terminal management.
- Any relocation of the terminals to the urban periphery should be accompanied by these institutional reforms—transparent revenue collection, depot requirements in route permits

and adequate public staffing—so that the new terminals do not reproduce the problems of the existing ones.

Implementing these recommendations would change the relationships among stakeholders and their influence over terminal management. The expected changes are illustrated using a reformed power–influence matrix and a potential Venn diagram.

- **Power–influence matrix after reform**

Figure 9.2 shows the power–influence matrix expected after the recommendations are implemented.

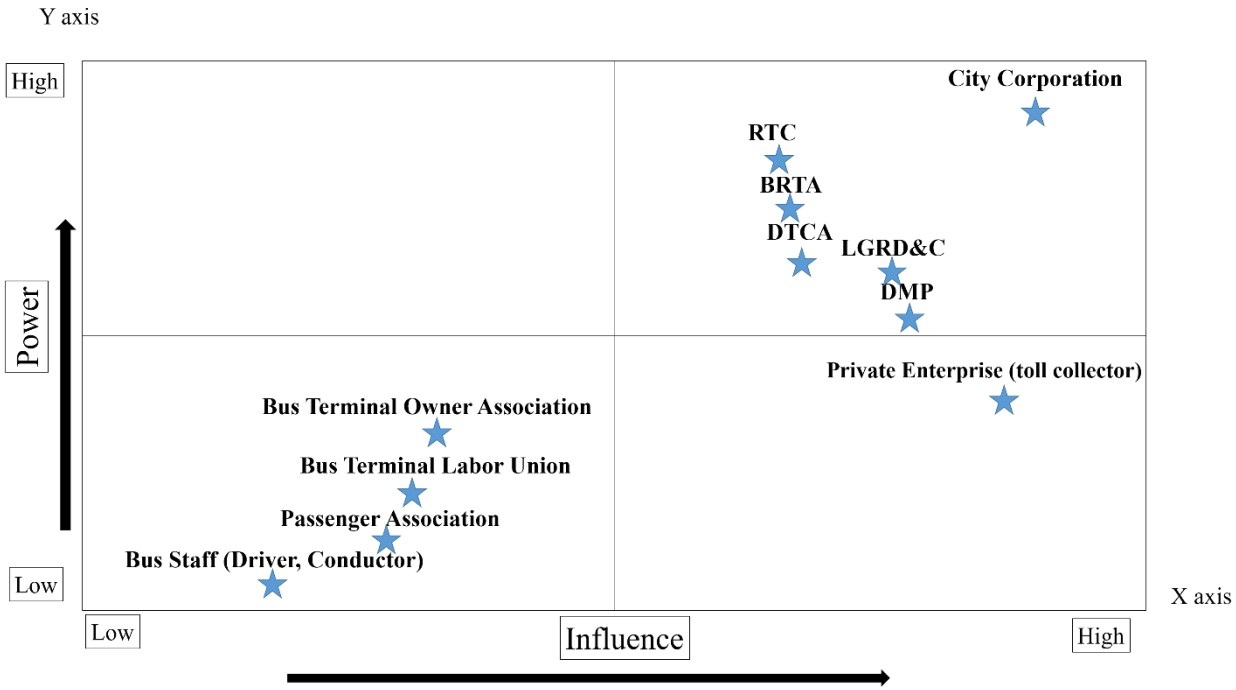


Figure 9.2: Power influence matrix after reform

In the reformed matrix, each stakeholder's influence is aligned with its formal power. The two city corporations, which hold the highest formal authority over the terminals, would also exercise the greatest influence. A private enterprise would replace the collectors; although it would have less formal power, it would play a defined role in terminal management. Other government agencies would also increase their influence. A passenger association would represent passengers' interests alongside the owners' associations and labor unions, but all associations and unions would have less influence and less power than at present.

- **Potential Venn diagram**

Figure 9.3 shows a potential Venn diagram of stakeholder relationships after reform, with the following features:

- As the principal authorities responsible for the terminals, the two city corporations should have the greatest influence, particularly over toll collection, relative to the labor unions, owners' associations and other stakeholders.
- The influence of local politicians should be removed from the terminals. Although the labor unions and owners' associations are organized and politically affiliated, their role in terminal management should be politically neutral.
- Because toll collection is large and complex, the authority should involve a private enterprise in the process. This enterprise should have the second-highest influence in terminal management, and its interaction with the Supervisor should be closer and more intensive than at present.
- A passenger association should work in the terminals regularly alongside the labor unions and owners' associations, raising the problems caused by deficiencies in functional arrangement. A system should also be introduced that allows passengers to choose their bus and go to the loading area unaided, so that companies no longer need linemen.
- Informal maintenance workers should be replaced by formally employed City Corporation staff, making terminal management more efficient and formal.
- BRTA and RTC should become stricter and build their capacity to control route permits, in order to manage the growing number of buses using the three terminals.
- DMP's role in ensuring the security of passengers and staff in the terminals should be strengthened.

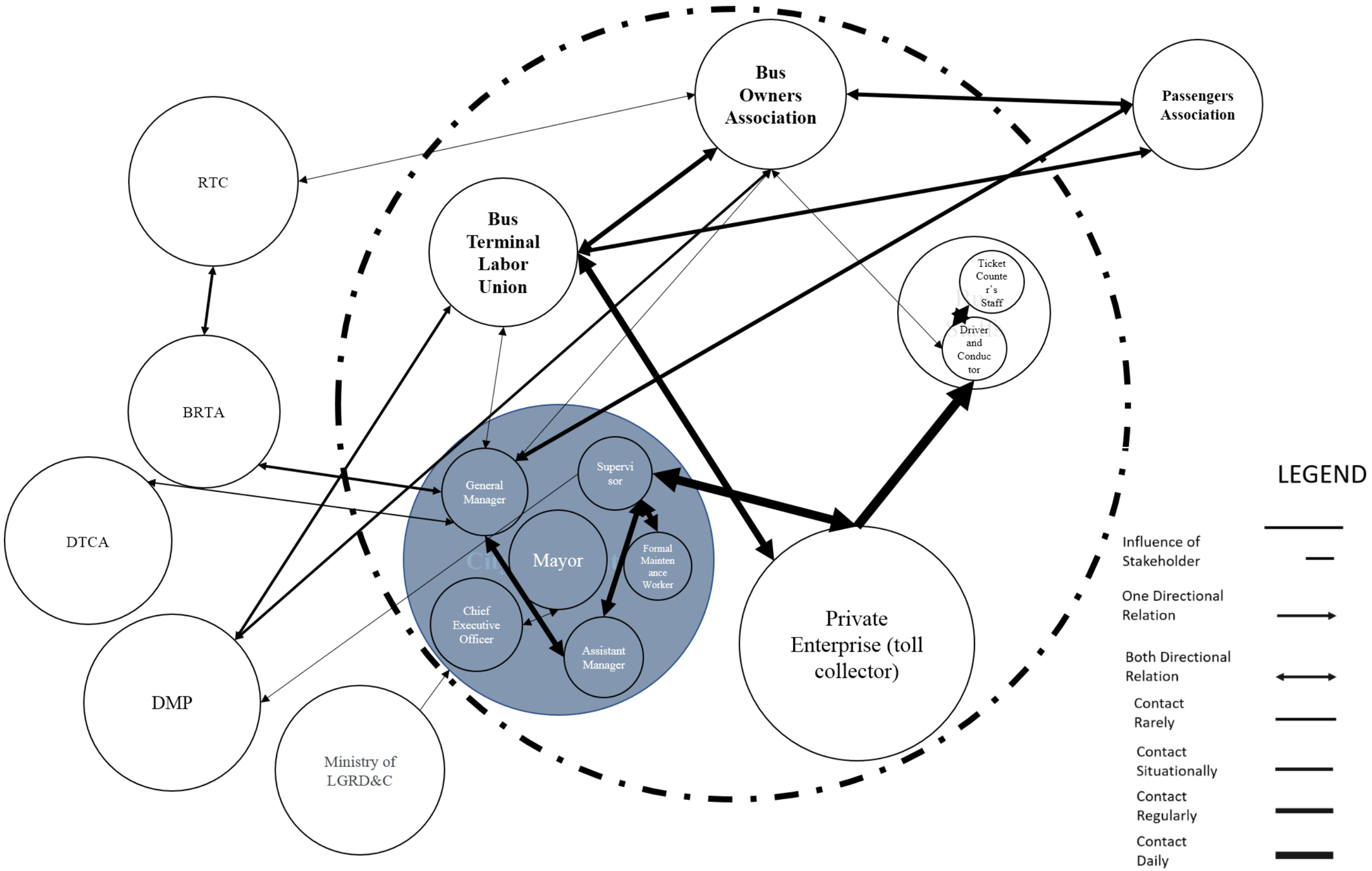


Figure 9.3: Potential Venn diagram of stakeholders of bus terminals

Chapter 10: Conclusion

The inter-district bus terminals of Dhaka are key elements of Bangladesh's national passenger transport system, yet all three operate ineffectively. This study has documented deficiencies in their functional arrangement—roadside loading and unloading, ticket counters outside the terminals, the use of terminals as depots and workshops, and under-used or absent amenities for passengers and crews—and has shown that these deficiencies are closely linked to the way the terminals are managed. Formal authority rests with the city corporations, but effective control is exercised by owners' associations, labor unions and their political patrons, while chronic under-staffing and informal toll collection leave the public authority unable to maintain the terminals. The relationships, influence and power of the stakeholders therefore need to be reformed, and the government needs to give urgent priority to the functional and managerial efficiency of the terminals.

The study's broader contribution is to show that the performance of transport infrastructure in rapidly growing cities cannot be understood from its physical design alone. In Dhaka's terminals, informal governance arrangements are inscribed in space: they determine where passengers board, where buses park and which facilities are used. Planning standards and physical upgrading are necessary but not sufficient; without institutional reform, new or relocated terminals are likely to reproduce the same pattern.

Several directions for further research follow. First, the qualitative evidence presented here could be complemented by quantitative measurement—for example, GPS and video data on bus dwell times and roadside stopping, or passenger surveys of terminal use—to estimate the congestion and welfare costs of informal terminal operation. Second, comparative research across South Asian cities, and with cities in Africa and Latin America where transport associations play similar roles, would show how far the mechanisms identified here are general. Third, the planned relocation of Dhaka's terminals offers an opportunity to study, before and after the move, whether changes in physical infrastructure alter the underlying governance arrangements. Finally, the persistence of terminal control across changes of government warrants closer study of how transport governance is embedded in the wider political settlement (Khan, 2010).

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Appendix

Appendix A: Checklist for Terminal Staff, Bus Staff and Terminal Amenities

Terminal Name	Parking Facility for Bus	Parking Facility for Other Transport	Entry and Exit		Arrival Lane	Departure Lane	Mixed Lane	Passenger Loading and Unloading Area	Ticket Counter		Information Desk Facility	Rest Room	Toilet Facility	Bus Washing and Clearing Facility	Emergency Repair Facility
			Separate	Merge					Permanent	Temporary					
Sayedabad	Partly	Yes	Yes	No	Yes	Yes	n.r.	Partly	Yes	Yes	No	No	Yes	Partly	Partly
Mohakhali	Partly	Partly	Yes	No	Yes	No	n.r.	Partly	Yes	Yes	Partly	No	Yes	Partly	Partly
Gabtoli	No	No	Yes	No	Yes	Yes	n.r.	No	Yes	Yes	No	No	Yes	Partly	No

Key: Yes = present; Partly = present but insufficient or not used for its intended purpose; No = absent; n.r. = not recorded. Source: observation survey, November 2018–February 2019; entries compiled from the findings reported in Sections 5.1–5.3.

Notes: Sayedabad – 500 bus parking spaces, below demand; bus bays not used for boarding; one washing ramp; informal repair shops only. Mohakhali – 128 parking spaces for about 800 buses; no departure lane; loading lane occupied by one operator; information desk not always staffed; designated repair area unused. Gabtoli – no designated bus parking or loading area; feeder vehicles park on the highway; washing area too small; repairs carried out in the parking area.

Appendix B: Checklist for Passenger Amenities

Terminal Name	Passenger Shed	Waiting Room	Toilet	Locker Room	Shops	Medical Center	Mosque	Telephone Room	Information Center
Sayedabad	Partly	Partly	Yes	No	Yes	No	n.r.	n.r.	No
Mohakhali	Yes	Partly	Yes	n.r.	Yes	n.r.	n.r.	n.r.	Partly
Gabtoli	Partly	Partly	Yes	n.r.	Yes	n.r.	n.r.	n.r.	No

Key: as for Appendix A. Source: observation survey, November 2018–February 2019; entries compiled from the findings reported in Chapters 5 and 6.

Notes: Sayedabad – passenger sheds little used; waiting room occupied by vendors and staff; no locker room, medical center or information center (Section 6.2.3). Mohakhali – waiting-room seating inadequate at peak times; information center not always staffed. Gabtoli – passenger sheds overcrowded before national holidays; waiting room in good condition but underused

Appendix C

Questionnaire for Key Informant Interview

Bangladesh University of Engineering and Technology

Department of Urban and Regional Planning

A STUDY ON THE PROBLEMS OF BUS TERMINALS IN DHAKA CITY: FUNCTIONAL ARRANGEMENT AND MANAGERIAL ISSUE

We are the students of Level IV, Term II of the Department of Urban and Regional Planning , Bangladesh University of Engineering and Technology are conducting a research on "A Study on the Problems of Bus Terminals in Dhaka city: Functional Arrangement and Managerial issue" To fulfill our study we need to interview experienced person. All information derived from this study will be used solely for academic purpose.

.

City corporation official

1. **Name:**
2. **Designation:**
3. **Office/ Work place:**
4. **How many employees of city corporation are working in the terminal?**

…….

5. List of the employee (formal):

……

……

……....

6. Is there any informal workers work in the terminal? If yes, then provide the list of the employee.

……

……

……

……..

7. How is the informal employee recruited?

……

……

8. What is the hierarchy of authority related to bus terminal in city corporation?

……

……..

9. What is the activity process of administrative side of city corporation?

……..

……..

……..

10. How is the relation between city corporation official and malik somiti/sromik somiti?

……...

……..

……

11. What types of problems occur in the terminal?

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……

……

12. How is the problem solved?

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……

……

13. Describe the parking problems of bus terminal.

……

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……

……

……

14. Why is the police box required in the terminal?

……

……

……

15. Is there any special maintenance worker for waiting room?

……

16. Why is the toilet in the waiting room different from others?

……

……...

17. Is there any designated bus washing area and charge for the usage of the area? Describe the process.

……

…….

18. Why is there no bus crew resting area in the terminal?

……

…….

……..

19. Is there any information desk facility? .....

19.1 If yes, what kind of information do they provide?

19.2 If no, then why?

……..

……..

20. What is the number of ticket counters in the terminal?

……

21. How is the rate determined?

…….

22. Are shops in the terminal enlisted? What is the process of renting shops in the terminal?

……

…….

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Labor Union

1. **Name:**
2. **Designation:**
3. **Office/ Work place:**
4. **How many labors of union are working in the terminal?**

…….

5. List of the labor who work in the terminal?

……

……

……....

6. What is the role of labor union in the management system of terminal?

……

……

……

7. What is the hierarchy between the labors of the terminal?

……

……

8. Due to insufficiency of formal stuffs, how is the terminal managed?

……

……..

9. How is the parking management system worked in the terminal?

……..

……..

……..

10. How is the relation among city corporation official and owners association and labor union?

……...

……..

……

11. What types of problems occur in the terminal?

……

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……

……....

12. How is the problem solved?

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……..

……..

13. How are the Kulis worked in terminal?

…….

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…….....

14. Is there any designated bus washing area and charge for the usage of the area?

……

……..

……

15. How are bus washing and repairing system worked?

……

……

……

16. Is there any special facilities for the labor of the terminal apart from the terminal stuffs and passengers, if yes then what are the facilities?

…….

……...

Signature of Interviewee: …….

Appendix C

Questionnaire for Key Informant Interview

Bangladesh University of Engineering and Technology

Department of Urban and Regional Planning

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We are the students of Level IV, Term II of the Department of Urban and Regional Planning , Bangladesh University of Engineering and Technology are conducting a research on "A Study on the Problems of Bus Terminals in Dhaka city: Functional Arrangement and Managerial issue" To fulfill our study we need to interview experienced person. All information derived from this study will be used solely for academic purpose.

Bus Terminal Owners Association

1. **Name:**
2. **Designation:**
3. **Office/ Work place:**
4. **What is the role of the owners association in the operation and management of the terminal?**

……..……

……..……

……

5. List of the employees who are employed in the terminal by the owners association?

……

……

……....

6. How is the parking space determined for each bus? Is there any role of the owners in parking management?

……

……

……

7. How is the employee recruited under each owner of the bus?

……

……

8. How is the relation between labor and owners association?

…….

……..

9. How is a problem solved between labor and owners association?

……..

……..

……..

10. How is the relation between city corporation official and owners association?

……...

……..

……

11.Why are some ticket counters unused?

…….

……

12. Is their any formal ticketing system along side the ticket counter in terminal, if yes then why?

…….

…….

……..

……..

13. What are the alarming issues in the terminal from the point of view of bus owners association?

…….

…….

…….

…….

14. Is there any plan of the owners association regarding the complain and problem of the terminal?

……

……..

……

Signature of Interviewee: …….

Appendix C

Questionnaire for Key Informant Interview

Bangladesh University of Engineering and Technology

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Expertise

1. **Name:**
2. **Designation:**
3. **Office/ Work place:**
4. **What is your opinion regarding the management of bus terminal?**

……..

5. What is your opinion regarding functional system of bus terminal?

……..

6. How can this be solved effectively?

……

……

……

Appendix D

Questionnaire for Focus Group Discussion

Bangladesh University of Engineering and Technology

Department of Urban and Regional Planning

A STUDY ON THE PROBLEMS OF BUS TERMINALS IN DHAKA CITY: FUNCTIONAL ARRANGEMENT AND MANAGERIAL ISSUE

We are the students of Level IV, Term II of the Department of Urban and Regional Planning , Bangladesh University of Engineering and Technology are conducting a research on "A Study on the Problems of Bus Terminals in Dhaka city: Functional Arrangement and Managerial issue" All information derived from this study will be used solely for academic purpose

1. Which problem do you face mostly regarding bus terminal facilities?	
2. From where do you get in the bus after buying ticket?	
3. Which facilities do you use mostly?	
4. Are you satisfied with the facilities provided for you?	
5. Do you need any more facilities?	
6. What is your opinion regarding the management of bus terminal	
7.What is your opinion regarding functional system of bus terminal?	
Participants Name 1…….. 2……. 3……..	

4……...
5……...

Appendix E

Key Informant Interview

KII 1 City Corporation Official

It is the compilation of information which was collected from the interview with the city corporation official. As all the questions are not answered by the same person so same questionnaire was used for all officials to reduce the information gap.

Mayor and Chief executive officer are mainly approval authority of City Corporation. Assistant manager is the personnel who directly works in the terminal office and supervisor helps him to do his daily activities. They mainly collect tolls from buses, toilets, shops daily. The number of Clerk, electrician, security, plumber, cleaner are insufficient and sometimes workers are hired to run the bus terminal. City Corporation get 117000 tk. per day from Gabtoli bus terminal, average 65000 tk. per day from Sayedabad bus terminal and 50000 tk. from Mohakhali bus terminal. It is called "Khas collection"

There are other government organization like DMP, BRTA, RTC, DTCA, LGRD&C who are also involved in managerial activities of bus terminal. BRTA and RTC are working with the route permit and documents of buses.

Bus Owners Association of related terminal make highest influence in the management system of all three terminals though the DNCC and DSCC are the authority to manage the terminals.Bus Owners Association and Bus Terminal Labor Union influence the managerial and revenue collection process. As there was case against Rora enterprise which was the private authority of Sayedabad and Mohakhali Bus terminal in 2004-2005, no new private organization can be hired.

KII2 Bus Terminal Owner Association

It is the compilation of information which was collected from the interview with the members of the bus terminal owner association. As all the questions are not answered by the same person so same questionnaire was used for all the members to reduce the information gap.

Owners of the bus terminal buses are mainly the members of the association of the related terminal. If someone is not the member of this association, he can't use the terminal for parking their buses or loading passenger from there. Their main objectives are to prioritizing the owner's bus terminal related interest and to arrange meeting with government on a regular basis. The association members come to the office to take major decisions in the event of political occasion.

In the Motor Vehicle Ordinance, 1983 taking rest after every 5 hours long journey is necessary for drivers and conductors.

Besides owners of buses use the terminal as depot because most of the owners do the same. As there are so many bus companies for same route, competition among bus companies is noticeable. They have ticket counter space inside the terminal but they set up outside ticket counter to get more passengers.

KII 03 (Bus terminal Labor Union)

It is the compilation of information which was collected from the interview with the members of the bus terminal labor union. As all the questions are not answered by the same person so same questionnaire was used for all the members to reduce the information gap.

The main objective of the labor uniom is to improve the lifestyle of the labor. Driver, conductor, helper, engine mistri, electric mistri etc.are all members of these unions. Buses have to pay 40 tk each time to enter into the bus terminal but they pay about 400-500 tk. Due to the involvement of collector and kuli mojuri sorder, City Corporation has no authority over them as well as political influence is the main factor which causes the informal collection of toll

As there is excess demand for parking, internal road of the terminal is blocked to manage the parking. Parking on roadside disrupt the movement of other vehicles which results in traffic congestion. Resting room is not included on the design of three terminals. Since there is no rest room in the terminals, driver and conductor have to take rest in bus. Because of that, their physical and psychological health conditions are being declined

KII-04: Expert Opinion Regarding Managerial Issues of Three Inter-District Bus Terminals of Dhaka

KII-04 has been conducted with Dr. Shamsul Haque, Professor, Department of Civil Engineering, BUET. He suggested that number of bus companies should be less than five. Less number of bus

companies will control all buses which use the terminal so that number of ticket counters will be decreased according to his opinion. There will not be necessary of ticket counter at the outside of the terminal as well as tendency of idle parking in the terminal will be decreased. Passengers will come in the terminal to buy ticket and will use the facilities provided in the terminal. He also added that management system should be better in an integrated way that all associations, city corporation and government organization work coordinately.

Appendix F

Coordination Schema

Objectives	Issue	Variables	Data type	Data level	Data source
To study existing condition and functional arrangement of bus terminals in Dhaka city	Terminal Staff, Bus Staff and Terminal Amenities	Location of Parking Facility for Bus	Qualitative	Nominal	Primary and Secondary data
		Location of Parking Facility for Other Transport			
		Location of Entry and Exit			
		Location of Arrival and Departure Lane			
		Location of Passenger Loading and Unloading Area			
		Location of Ticket Counter			
		Location of Information Desk Facility			
		Location of Rest Room			
		Location of Toilet Facility			
		Location of Bus Washing and Clearing Facility			
		Location Emergency Repair Facility			
	Passenger amenities	Location of Passenger Shed			
		Location of Waiting Room			
		Location of Toilet			
		Location of Locker Room			
		Location of Shops			
		Location of Mosque			
		Location of Medical Center			
		Location of Information Center			
To identify the stakeholders involved and study the managerial issue related to the problems of the bus terminals		Stakeholders' position		Ordinal	Primary data
		Level of Power			
		Level ofinfluence			
		Intensity of interaction			

Appendix G: Images of



Figure: Passenger loading outside the terminal



Figure: Passenger offloading in the exit ramp of flyover



Figure: Passenger Shed



Figure: Ticket Counter outside the terminal



Figure: Passenger offloading on the road



Figure: Buses using flyover ramp



Figure: Haphazard parking on exit point



Figure: Parking on Jonopod Mor

Appendix H: Images of



Figure: Unused Ticket Counter



Figure: Canteen



Figure: Water bucket using for washing facility



Figure: Poor condition of environment

Appendix I: Images of



Figure: Repair Facility



Figure: Ena bus using bus bay



Figure: Feeder service usage



Figure: List of labor in the terminal

Appendix j: Other Images



Figure: Focus Group Discussion



Figure: Focus Group Discussion



Figure: Gabtoli terminal owner association



Figure: Gabtoli terminal labor union office

Appendix j: Other Images



Figure: Outside counter



Figure: Resting inside bus



Figure:Traffic congestion outside the terminal



Figure: Uncleansed environment

Appendix K

Management activity by different stakeholders

City Corporation Official	Mayor	• Mayor is the head of the Organization • Approval authority for administrative, financial, policy & decision making matters
	The Chief Executive officer (Additional Secretrary)	• Executive Head of the Organization • the controlling authority of all departmental activities. • In exercising his power and discharging duties he will be responsible to the Mayor
	General Manager (Deputy Secretary)	• Revenue Collection from Bus Terminals • Cleaning the Terminals. • Ensure Security & Service to passengers. • Development/public works
	Assistant Manager	• Maintenance and repairing of amenities (lighting, toilet, waiting room facility, internal road etc.) • Collecting toll from buses, toilets and shops • Managing the problem (illegal hawker, internal road encroachment) • Keeping daily record of entry and exit of buses • Ensuring safety of passengers in the terminal

	Supervisor	• Helping Assistant Manager to perform his duty
	Clerk	• Keeping records, accounts, and undertake other routine administrative duties.
	Cleaner	• Cleaning the bus terminal premises and the terminal building
	Security	• Ensuring safety of passengers
	Electrician	• Ensuring effective lighting system in the terminal • Maintenance of electrical equipment in the terminal
	Plumber	• Ensuring effective maintenance of sewage and drainage system of terminal. • Maintenance of water supply system of the terminal

Bus Terminal Owner Association	Chairman	• Can preside the meeting of the management committee. • Can call emergency meeting when needs
	Vice Chairman	• Can preside the meeting of the management committee in case of the absence of the chairman.
	General Secretary	• Control over the employees of association • Manages all the management expenses

		• Administrative duty
	Assistant General Secretary	• Performs all the responsibilities of the secretary in the absence of him. • Provides assistance to secretary in the time of need.
	Treasurer	• Handles cash and deposited property of the association in custody • Keeps the details of the cash.

Bus Terminal Labor Union	President	• Can preside management meeting and executing meeting • Can take major decision in emergency situation • Communicating with the bus terminal owner association to solve problem
	Vice President	• Can arrange management meeting and executing meeting
	General Secretary	• Manages all the management expenses • Calls general meeting
	Joint Secretary	• Performs all the responsibilities of the secretary in the absence of him.

	Sports Secretary	• To arrange different types of sports activity for the labors of the terminal annually
	Treasurer	• Keeps the details of the cash of the union member • Reserves all the deeds of the bank account • Submits all the receipts of the union
	Officer Secretary	• Keeping records, accounts, and undertake other routine administrative duties.
	Publication Secretary	• Maintaining diaries and arranging appointments. • Typing, preparing reports, leaflets, poster
	Executive member	• Coordinate to make a decision
	Bus Staff	• To operate the bus • Conductor wash the buses and do minor repair activities

Organization	DMP	• To control traffic around the bus terminal • To control illegal social activity inside the bus terminal • Checking passenger for carrying illegal goods

	BRTA	• Managing and monitoring bus terminal • Checking License and fitness of the buses
	RTC	• To provide route permit for new terminal buses
	DTCA	• To coordinate transport system of Dhaka including the bus terminal
	LGRD&C	• To appoint employees of bus terminal